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Numerical simulations of a rotor in confined areas including the presence of wind

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ABSTRACT

This work explores the high fidelity numerical aerodynamic computations of a helicopter rotor placed in various confined areas including the presence of wind. Two numerical methods, fully blade resolved CFD and the actuator line method (ALM), are compared against each other and experimental data in terms of global rotor loads including thrust, torque, pitching and rolling moments. A detailed assessment of the two methods outside-of-ground effect (OGE) is performed covering blade loading, rotor inflow, wake geometry and fuselage surface pressure coefficient. Excellent agreement and consistency are found between the two numerical methods. Both are able to accurately predict the trends of the experimental rotor in most cases. However, some conditions remain challenging from either a modelling or computational standpoint. The overall agreement of the two numerical methods with the experimental data demonstrates their capabilities in accurately predicting rotor flows in challenging confined areas and the appropriateness of the ALM to replace more costly fully blade resolved simulations.

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1. Introduction

Due to its hovering and manoeuvring capabilities, the helicopter is currently one of the most versatile flying machines in operation for air transport. This aircraft is capable of flight in challenging conditions such as in rescue scenarios over variable terrain, near city sky-rises, landing atop buildings or even operating in the close vicinity of naval vessels. All these scenarios involve complex geometries that can interact with the flowfield created by the helicopter rotor in addition to the already complex flowfield the obstacles produce by themselves in presence of wind. The disturbed flowfield has a strong influence on the performances and handling characteristics of the helicopter.

A well-studied example of handling degradation is the operation of a helicopter at sea performing a landing manoeuvre near a ship as described by Hoencamp et al. [1] from a pilot's perspective. This problem has also been documented in multiple experimental [2–5] and numerical [6–10] studies on numerous specific ship and helicopter geometries.

In addition to this very complex type of interaction, there is also an interest in studying the performances of helicopters in simpler settings such as in "confined areas" (i.e. where the helicopter sees its displacement restricted in at least one direction due to terrain or obstacles). This particular issue was the focus of the GARTEUR Action Group project HC/AG-22 [11] which produced multiple experimental databases that pertain to the analysis of small scale helicopter models in interaction with different obstacles [12–15] and the effects of the helicopter on the obstacle surface loads. The analysis of rotors in confined areas is also a field of interest for Micro Air Vehicles (MAV) which has been investigated both experimentally and numerically [16,17]. One motivation behind such studies is to gain phenomenological understanding of the operation of helicopters on simplified geometries. In addition, these simplified geometries are well suited for Computational Fluid Dynamics (CFD) code validation.

Simulating the aerodynamic performances of helicopters and rotors in complex environments can be done by a multitude of techniques. Simpler cases in-ground-effect or in advancing flight could be simulated through Blade Element Theory (BET) and Blade Element Momentum Theory (BEMT) [18] approaches given proper modelling considerations and calibration. As the level of complexity of the flowfield increases, these low fidelity models quickly become unable to predict rotor performances in these situations and increasingly rely on empirical laws derived from experimental

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Nomenclature

A_{disk}	Area of the rotor disk	m^2	
AR	Aspect Ratio, R/c_{ref}		
C_d	Blade section 2D Drag coefficient		
C_l	Blade section 2D Lift coefficient		
C_{Mx}, C_{My}	Rotor in plane moment coefficients, $M/(\rho_\infty V_{tip}^2 A_{disk} R)$		
$\overline{C_P}$	Average pressure coefficient, $(\overline{p} - p_\infty)/(0.5\rho_\infty V_{ind}^2)$		
C_Q	Rotor torque coefficient, $Q/(\rho_\infty V_{tip}^2 A_{disk} R)$		
C_T	Rotor thrust coefficient, $T/(\rho_\infty V_{tip}^2 A_{disk})$		
c	Local chord	m	
c_{ref}	Reference chord	m	
D	Rotor diameter	m	
$\mathbf{f_{alm}}$	Volumetric body force of the ALM		
FoM	Figure of merit, ${C_T}^{1.5}/(\sqrt{2}C_Q)$		
g	Gaussian kernel		
M	Mach number		
N_b	Number of blades		
R	Rotor radius	m	
Re	Reynolds number, $U_{rel}c/\nu$		
r', z'	Radial and axial rotor coordinate normalised by R		
$\mathbf{U_P}$	Perpendicular velocity to the rotor disk	m/s	
$\mathbf{U_{rel}}$	Relative velocity a blade section sees	m/s	
$\mathbf{U_s}$	Sampled velocity at kernel centre	m/s	
$\mathbf{U_T}$	Tangential velocity to the rotor disk	m/s	
U_∞	Freestream velocity	m/s	
V_{ind}	Hover induced velocity, $V_{tip}\sqrt{C_T/2}$		
V_{tip}	Velocity of the blade tip	m/s	
(x, y, z)	Rotor reference coordinate system		
(X, Y, Z)	Absolute reference coordinate system		
α_{rel}	Relative angle of attack of a blade section	deg	
ΔD	Blade segment drag force	N	
ΔL	Blade segment lift force	N	
ΔR	Blade segment radial length	m	
ϵ	Gaussian smearing width parameter		
Θ_{75}	Collective pitch at 75% blade span	deg	
μ	Advance ratio, U_∞/V_{tip}		
σ	Rotor solidity, $N_b c_{ref}/(\pi R)$		
$\Psi_{C_{Mtot}}$	Orientation of the total in plane moment coefficient	deg	
Ω	Rotor rotational speed	rpm	

data or higher fidelity models, such as CFD, for their calibration. Increasing in level of fidelity we find singularity based methods such as Vortex Lattice Methods (VLM) [19–22] and Panel Methods (PM) [12] which can handle simple geometries in an inviscid way. In recent years, these two methods have been coupled to Vortex Particle Methods (VPM) for the evaluation of the rotor wake, including viscous effects, and possible interaction with geometries such as a ground, box and even ships [10,23–25]. Finally, mesh-based CFD methods have been used extensively for the simulation of rotors ranging from relatively simpler cases of ground effect in advancing flight cases through increasingly complex environment such as the fully coupled rotor-ship dynamic interface [6,26,27]. These high fidelity methods usually offer better predictive capabilities as a result of a more complete description of the flowfield through the modelled equations and the explicit inclusion of geometries within their computational framework. However, this increased level of fidelity and accuracy comes at the cost of additional computational time.

A common trend in the study of helicopters and rotor in complex environment within a high fidelity framework is to use rotor replacement techniques. These simplifications remove the geometry of the rotor blades and thus their associated meshes. The compute time of the simulation is significantly reduced by the economy in mesh size but comes as a trade-off in accuracy, however. The effect of the rotor is replaced by momentum source terms in the (Unsteady) Reynolds-Averaged-Navier-Stokes ((U)-RANS) equations and exert force on the flowfield resulting in similar flow characteristics. The simplest form would be an Actuator Disk (AD) [28–31] which averages the rotor blade loads azimuthally, and sometimes radially. This approach yields a good prediction in average surface loads on a fuselage but loses the time accuracy and the distinct tip vortex wake. The AD has recently been used by Shi et al. [8,9,32] in the study of the helicopter/ship dynamic interface for a case with two rotors near a ship landing deck as well as in an investigation of the influence of passive/active flow control on the dynamic interface. An evolution of the AD would be to use an Unsteady Actuator Disk (UAD) as introduces by Boyd [33] with recent developments and different modelling strategies presented in Barakos et al. [34]. Oruc et al. [7,35] used the model in the study of the interaction between a rotor and a Ship Frigate in a simulation coupled with a flight dynamic model whereas Chirico et al. [36] used it to investigate the interaction between a rotor and a generic box-shaped obstacle. A further evolution is the Actuator Line Method (ALM), as introduced by Sørensen and Shen [37] in the Wind Energy sector, that has been one of the most used computational method for horizontal axis wind turbines in the past two decades but is seldom used in the helicopter community. The ALM represents the rotor blades as rotating lines of momentum source terms whose magnitudes are computed using a BET approach coupled to the 3D RANS flowfield. This method, unlike the AD, produces distinct tip vortices. A modified ALM was used by Alpman et al. [38] in addition to a helicopter flight dynamic model for the computation of rotor load time histories for a rotor placed behind a hangar with incoming wind as well as a rotor behind a ship frigate to evaluate different coupling procedures. Forsythe et al. [39] used a swept version of the ALM to achieve larger time steps towards applying it to a helicopter landing on a ship. Delorme et al. [40] used the ALM to compute rotor noise in a Large Eddy Simulation (LES) framework through the use of a Ffowcs Williams-Hawkings acoustics analogy model. Then the ALM has recently been validated in hovering conditions on the S-76 rotor by Merabet and Laurendeau [41]. Finally, a further variation of the ALM would be to consider a chord-wise distribution of the sectional force with either constant loading or with a distribution mimicking the blade surface pressure coefficient and possibly skin friction coefficient. This Actuator Surface Method (ASM), as introduced by Shen and Sørensen [42] and Sibuet-Watters and Masson [43–45] in the wind turbine industry, was recently used by Linton et al. [46–48] to simulate a helicopter rotor in hover, forward flight, in a tandem configuration and in interaction with a ship wake for the evaluation of the dynamic interface.

An outstanding issue in the study of helicopters in complex or confined environment, in particular for the study of rotor-ship dynamic interface, is the lack of proper validation data for CFD purposes. For the specific case of rotor-ship dynamic interface, several experimental studies have been performed in the past [2–4,49–52], with some only using a ship geometry without a rotor whereas others, that did include a rotor, contained complex and or proprietary ship/fuselage geometries. In the context of CFD validation of the complex flowfield of a helicopter in a confined area, it is desirable to simplify the geometry as much as possible. Through the GARTEUR AG22 project, the experimental database of Zagaglia

et al. [53] achieves this goal. The obstacle is a well-defined parallelepiped with sizes of the order of 1 to 3 rotor radius. The fuselage is simple, minimalist and does not contain small intricacies whereas the rotor is simplified through the use of well-defined airfoil profiles, fixed collective, no cyclic variation and a rigid rotor. Although these modifications produce a helicopter that is untrimmed and do not reproduce accurately the dynamics of a real helicopter, the authors of the study believe that the main features of the wake are still representative of a real helicopter. Furthermore, on a CFD simulation standpoint, it is simpler to model a helicopter rotor with a fixed geometry that is known *a priori* as it eliminates possible errors and modelling that pertain to blade motions. The study of Zagaglia et al. covers multiple flight conditions in increasing level of complexity ranging from hovering in and out-of-ground-effect, with and without wind (advancing flight) and with the presence of a box-shaped obstacle with the rotor placed at different positions relative to it. Measurements of the rotor thrust, torque, rolling and pitching moments, Particle Image Velocimetry (PIV) and pressure loads on the obstacle are performed to form the database. However, only the rotor loads are reported in their published study, which is plentiful for an initial effort in validation of computed rotor loads through CFD. Consequently, this work uses the database as a validation test case.

For the numerical simulation tools, two methods are considered, starting with a fully Blade-Resolved (B-R) CFD using the Overset Mesh technique within a U-RANS framework. This high fidelity simulation will allow an assessment of the current state of the art for this class of simulation tool against a recently published dataset as well as provide a benchmark for the rotor replacement technique. The second method considered is the ALM and comes as an evolution of actuator disks, which are typically used for this kind of simulations. In fact, as reported above, higher fidelity rotor replacement techniques (i.e. ALM and ASM) are starting to gain attention in the helicopter community especially with the aim of tackling complex problems of rotor in confined areas or the rotor-ship dynamic interface. The ALM will therefore be benchmarked against the B-R solutions on identical meshes and simulation templates in order to assess the performance of the method for rotors in confined areas.

In this work, a helicopter rotor is simulated in multiple flight conditions starting with a comprehensive assessment of the hovering and forward flight performances outside of ground effect including detailed performance coefficients evaluation, blade loading, inflow, wake geometry and fuselage surface pressure. Then, for both these conditions, the in-ground-effect (IGE) configuration is studied. It is followed by the addition of the obstacle placed upstream of the rotor with the latter performing a vertical sweep to assess the effects of the obstacle on the ground effect.

2. Numerical methodology

The flow solver used in this work is Star-CCM+ 12.06 [54] which is a general purpose finite volume, unstructured, cell centred, multi-physics flow solver. All computations are done with a segregated approach using the SIMPLE [55] algorithm, including compressibility corrections, with an unsteady dual time-stepping scheme with 2nd order accuracy in both time and space. The time step used for all simulations in this work corresponds to an equivalent of 1° of azimuthal rotation. All the simulations in this work are solved with the $k-\omega$ SST turbulence model [56]. The ALM adds momentum and energy source terms to the U-RANS equations to simulate the effects of the rotor. The implementation is performed through the User Coding framework of Star-CCM+. Both the B-R and ALM solutions use the same simulation template in terms of numerical setup and meshes for a fair comparison between the two methods. It should be noted that the ALM is considerably less sensitive to mesh coarsening than the B-R and would produce accurate results in terms of performance coefficients on coarser meshes as explored in Ref. [41] for hovering conditions. The flowfield, however, would appear somewhat different as a result of the loss in mesh definition.

Blade resolved simulations are modelled through the Overset mesh technique. The rotating blades are individually meshed and interfaced with a static background grid for communication between the different meshes in the hierarchy with the interpolation between the different levels being done with an inverse-distance weighted approach. The mesh regions around the blades are either cylindrical or prismatic shaped and extend $\sim 1.25c_{ref}$ and $\sim 0.375c_{ref}$ away from the blade geometry respectively. The larger cylinder overset mesh region is used for isolated rotor simulations whereas the tighter prismatic mesh region is used when a fuselage is introduced in order for the blade mesh to clear the additional geometry as illustrated in Fig. 1. For both cases, the cell mesh size at the surface of the overset regions approximately matches the background grid.

2.1. ALM modelling

By removing the blades geometry and their associated meshes, the ALM replaces the physical representation of the rotor by source terms in the fluid flow equations. These equations are summarised in their generic integral form, omitting any turbulence model, by Eqs. (1)-(3), as per Ref. [54]:

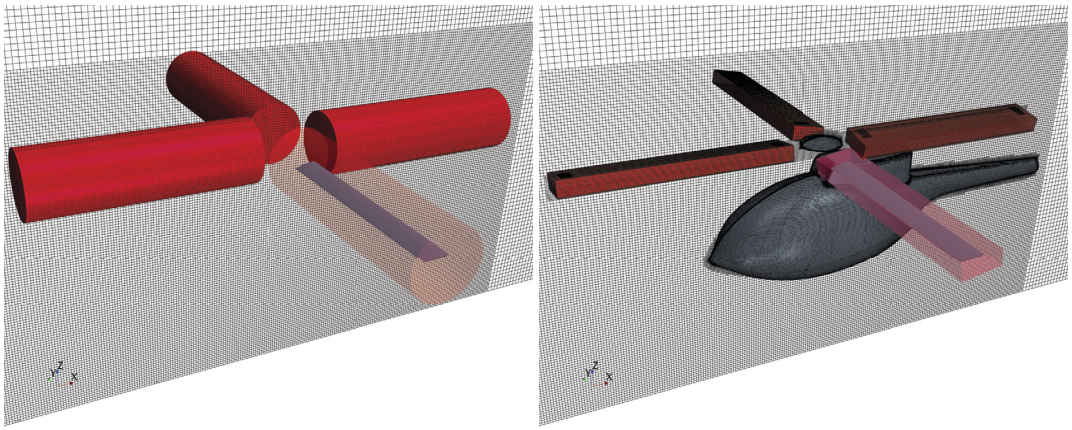
$$\frac{\partial}{\partial t}\int_V \rho\, dV + \oint_A \rho \mathbf{v}\cdot d\mathbf{a} = 0 \tag{1}$$

$$\frac{\partial}{\partial t}\int_V \rho\mathbf{v}\, dV + \oint_A \rho\mathbf{v}\otimes\mathbf{v}\cdot d\mathbf{a} + \oint_A p\mathbf{I}\cdot d\mathbf{a} = \oint_A \mathbf{T}\cdot d\mathbf{a} - \int_V \mathbf{f_{alm}}\, dV \tag{2}$$

$$\frac{\partial}{\partial t}\int_V \rho E\, dV + \oint_A \rho H\mathbf{v}\cdot d\mathbf{a} + \oint_A \mathbf{q}\cdot d\mathbf{a} = \oint_A \mathbf{T}\cdot\mathbf{v}\, d\mathbf{a} - \int_V \mathbf{f_{alm}}\cdot\mathbf{v}\, dV \tag{3}$$

where V is the control volume of a cell and $\mathbf{a}$ its outward area vector. Flow properties are represented by the density ρ, the velocity vector $\mathbf{v}$, the pressure p, the viscous stress tensor $\mathbf{T}$, the heat flux $\mathbf{q}$, and the total energy and enthalpy E and H, respectively. The body force needed for the modelling of the ALM is expressed by $\mathbf{f_{alm}}$ which is treated as any other volumetric body force. The negative signs indicate that the forces to be applied on the flowfield are of equal strength, but opposite direction of the rotor forces. The removal of the blade meshes produces substantial savings in terms of grid size yielding up to 3-4x speed-up in compute time as well as faster set-up time and increased parallel scalability [41].

Through a BET approach, the $\mathbf{f_{alm}}$ source term is modelled by considering the individual rotor blades as rotating lines that are discretised in sections along their span axis located on the quarter chord of the profile of the blade. In this work, each blade is discretised in 65 span-wise sections following a normalised hyperbolic tangent function [41]. This allows for a smooth and controllable clustering of the sections near the blade tip where the increased resolution better captures the stronger flow variation and loading gradient in this region while keeping the total amount of ALM sections low. Each section is considered radially independent from its neighbours and has its velocity sampled along its centre. Through



(a) Isolated rotor (b) Rotor with fuselage

Fig. 1. Overset mesh template.

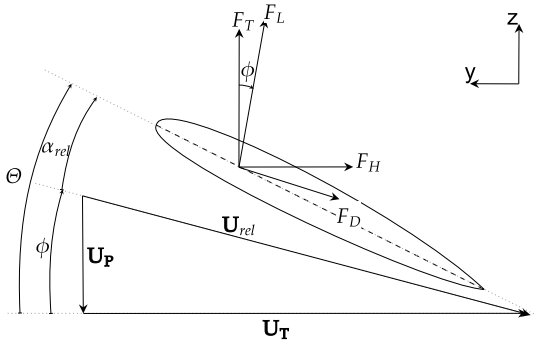


Fig. 2. Blade element representation of a blade section from [41].

the use of the integral velocity sampling [39,57], the sampled velocity $\mathbf{U_s}$ represents only the wake induced velocity and is void of any bound circulation induced velocity. As per BET, a local effective freestream velocity $\mathbf{U_{rel}}$ and angle of attack α_{rel} is reconstructed through Eqs. (4)-(7) and illustrated in Fig. 2.

$$\mathbf{U_T} = \Omega(r \cdot R) - \mathbf{U_{s,y}} \tag{4}$$

$$\mathbf{U_P} = -\mathbf{U_{s,z}} \tag{5}$$

$$\mathbf{U_{rel}} = \mathbf{U_P} + \mathbf{U_T} \tag{6}$$

$$\phi = \arctan\left(U_P / U_T\right) \tag{7}$$

With the freestream velocity and angle of attack computed, the ALM uses pre-computed values from a 2D viscous database to identify local lift, drag and moment coefficients. For the sake of consistency, the 2D polars are computed using the same numerical setup as the 3D blade resolved cases with a second order spatial stencil for the convective terms in a steady state formulation along with the k-w SST model as a turbulence closure. The lift and drag forces of Eqs. (8) and (9) are then projected onto the flowfield. The database is evaluated on 6 blade sections along the span with local Mach and Reynolds numbers over an angle of attack range of 0° to 20° as a result of the symmetric airfoil geometry.

$$\Delta L = \frac{1}{2}\rho \|\mathbf{U_{rel}}\|^2 C_l(\alpha_{\text{rel}}, \text{Re}, M) c \Delta R \tag{8}$$

$$\Delta D = \frac{1}{2}\rho \|\mathbf{U_{rel}}\|^2 C_d(\alpha_{\text{rel}}, \text{Re}, M) c \Delta R \tag{9}$$

The computed lift and drag forces from a particular ALM segment are projected into the flowfield through a Gaussian kernel, represented in Eq. (10), that spreads the force radially from the collocation point over several grid cells. The extent of the spreading is controlled by the ϵ term which is typically scaled by the local chord with values around $\epsilon \sim 0.25c$ showing better predictive agreement as explored in Refs. [58–61]. For the present study, the value of $\epsilon = 0.225c$ is used as it produced marginally better results than $\epsilon = 0.25c$. The ALM force source term $\mathbf{f_{alm}}$ is then reconstructed by summing every individual ALM segment contribution as in Eq. (11):

$$g(x, y, z) = \frac{1}{\epsilon^3 \pi^{\frac{3}{2}}} \times \exp\left(-\frac{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}{\epsilon^2}\right) \tag{10}$$

$$\mathbf{f_{alm}} = \sum_i \cdot g_i \cdot \left(\Delta L_i \mathbf{e_{L,i}} + \Delta D_i \mathbf{e_{D,i}}\right) \tag{11}$$

The source term used in this work is also radially truncated to respect the geometrical limits of the span of the rotor blade and normalised. As a result of the use of a smaller kernel size, no tip loss function is applied towards the blade tips. More details about the method and its adaptation to helicopter rotors can be found in Ref. [41].

3. Test cases and geometry

3.1. Experimental test case description

The experimental study of Zagaglia et al. [53] is used to validate the performances of the two numerical methods for a helicopter in confined areas. The main characteristics of the helicopter rotor are summarised in Table 1. In addition, the experimental setup featured a reduced-scale fuselage based on a MD500 geometry. The motor and load measuring devices were embedded inside the fuselage which was held in place by a rear sting exiting the fuselage tail boom. The internal balance was fixed in such a way that it only measured the rotor loads, without the forces acting on the fuselage. Multiple configurations were studied including the presence of a box-shaped obstacle geometry and wind. The obstacle had dimensions of 1 m x 0.45 m x 0.8 m (W x H x L) whereas the Politecnico di Milano wind tunnel test section was 13.84 m x 3.84 m x 38 m (W x H x L). The wind was blowing at about 5.07 m/s which corresponds to an advance ratio of $\mu = 0.05$. As the ground was stationary, a boundary layer (BL) at the centre of

Table 1
Model rotor characteristics.

Rotor		Blade	
Nb. Blades (N_b)	4	Airfoil Profile, constant	NACA0012
Rotor Radius (R)	0.375 m	Reference Chord (c_{ref})	0.032 m
Collective, fixed (Θ_{75})	10°	Linear Twist	0°
Aspect Ratio (AR)	11.72	Tip Mach number (M_{tip})	0.3
Solidity (σ)	0.10865	Tip Reynolds number (Re_{tip})	220000
Rotational Speed (Ω)	2580 RPM		

the test section is present with a thickness of 0.18 m. This height corresponds to 40% of the obstacle height and a rotor normalised height of $Z/R = 0.48$. As such, the boundary layer influences the flowfield around the obstacle, but marginally affects the rotor as it was never placed lower than $Z/R = 1.0$ (double the BL height). Multiple tests were performed both in the presence ($\mu = 0.05$) and absence ($\mu = 0.0$) of wind with the rotor isolated from the obstacle (T0), placed above the obstacle (T1), performing a horizontal sweep above an edge of the obstacle (T2) and performing a vertical ground effect sweep centred (T3) and offset (T4) behind the obstacle.

3.2. Numerical setup, meshes and simplifications

For the present study, the test cases of interest are: hover in-ground-effect T0-μ=0.0, advancing flight in-ground-effect T0-μ=0.05, hovering in-ground-effect with obstacle T3-μ=0.0 and in windy conditions in-ground-effect with obstacle T3-μ=0.05. As such, different modelling strategies are used to accommodate the different setups and flow regimes. Most simulations only consider an isolated rotor with a root cut-out at 20% span. In cases where a fuselage is present, the main body shell is modelled along with the hub cap and all surfaces have 25 cells in their BL mesh and a $y^+_{wall} \sim 1.0$. No hub or linkages, rear sting and support structure are considered. For cases without the obstacle (T0, with and without wind), the floor is assumed to be inviscid as the modelling of the floor BL was tested and did not impact the rotor performance coefficients. For cases with the obstacle, both it and the floor have their boundary layer meshed and modelled with a height that captures the wind tunnel BL and a near-wall spacing that ensures $y^+_{wall} \sim 1.0$. The modelling of the obstacle and wind tunnel floor BL allowed a qualitative assessment between the present T3-μ=0.05 test case, with the rotor removed, and both the experimental work of Martinuzzi and Tropea [62] and numerical work of Hwang and Yang [63] and Lim et al. [64]. All numerical simulations are performed in a computational domain that represents the PoliMi wind tunnel test section. Mesh refinement zones are placed in the near wake region of the rotor. An additional volumetric refinement zone is added around the obstacle when present. B-R blade meshes have a near wall spacing of $5e^{-5}c$ and 35 BL cells which ensures a $y^+_{wall} \leq 1.0$. Fig. 3 illustrates an example of the mesh template used for the flow domain and B-R blade. Table 2 presents the total mesh size for both methods with and without the obstacle in a configuration near the ground and one in outside-of-ground effect (OGE). The decrease in mesh size for configurations closer to the ground is due to the reduction in size of the wake refinement region as it extends from a fixed height above the rotor to the ground. In all present cases, the ALM has mesh sizes roughly half of the B-R.

Experimental data points were sampled over a 5 seconds averaging window that followed a 10 seconds flow development/stabilisation period. This time interval corresponds to a total of 645 rotor revolutions, which represents an exorbitant amount of computational resources in a U-RANS framework if the numerical simulations were to match the number of rotations. Therefore, simpler cases (i.e. T0) are run for a total of 30 to 45 revolutions with the performance values averaged over the last 10 revolutions. Harder cases with the obstacle are run over a longer period of time ranging from 45 up to a maximum of 100 revolutions with the averaging performed over the last 30 revolutions in order to capture the greater flow unsteadiness effect over the rotor performances.

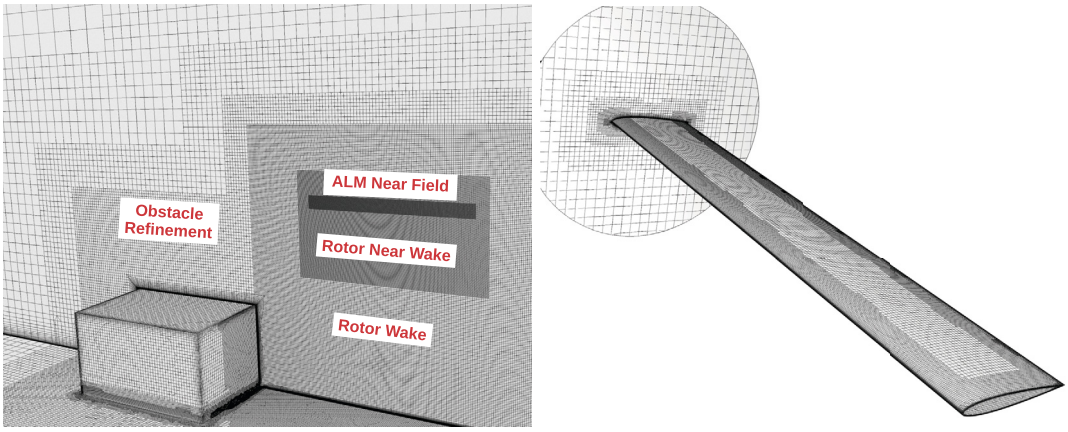
Table 2
Mesh cell count in different configurations.

Config.	Z/R	Cell count (M) B-R	Cell count (M) ALM
T0	1	25.6	12.2
	4	30.0	16.7
T3	1	28.9	13.9
	4	32.9	18.0

4. Results

4.1. Reference conditions OGE

First, the results of the reference cases for the two numerical methods with no obstacle and out-of-ground-effect (T0, Z/R=4) in both wind conditions are assessed. These reference values will be used to normalise the results from other test conditions in the presence of the ground and the obstacle. Numerical results for an isolated rotor with no fuselage, along with experimental data, of the integrated performance coefficients, are presented in Tables 3 and 4 for the wind-off and wind-on cases respectively. For the case in hover, the thrust, torque and figure of merit of both numerical methods are in good agreement with the experimental data with relative errors of the order of one percent. The values of the moment coefficients in both axes for both numerical methods are almost negligible (less than 0.15% of the OGE torque value) which is expected due to the axisymmetric nature of the flowfield in hover when only the main rotor is modelled. The slightly higher value observed for the experimental data could be explained by the presence of the fuselage in the physical experiment which breaks the axisymmetric hypothesis. Due to their small magnitudes, the relative errors of the moment coefficients is over-inflated and should not be considered. Next, the addition of wind causes an increase in the thrust, reduction in torque and hence an improved figure of merit as reported in Table 4. Both numerical methods show consistent trends when compared to the experimental data with an increase of 4% to 5% in thrust when compared to the hovering case. But the B-R slightly over-predicts the thrust and under-predicts the torque thus resulting in a larger FoM error. The two numerical methods correctly capture the signs of the moments in both axes, indicating a tendency of the rotor to pitch up in the rotor y-axis and a negative rolling moment towards the port side of the helicopter in the x-axis. The magnitudes of the individual axis components present a large error when compared to the experimental data, especially for the rolling moment. However, when looking at the total moment magnitude and the angle it makes in the rotor plane, the magnitude of both methods is better represented when compared to the experimental data



(a) Domain mesh T3 (b) Blade surface mesh

Fig. 3. Computational mesh template.

Table 3
Rotor performance coefficients OGE $\mu = 0.00$.

	Exp.	B-R	Δ	ALM	Δ
C_T^{oge}	7.268e-3	7.171e-3	-1.3%	7.193e-3	-1.0%
C_Q^{oge}	7.804e-4	7.714e-4	-1.1%	7.742e-4	-0.8%
FoM^{oge}	0.561	0.557	-0.9%	0.557	-0.8%
C_{Mx}^{oge}/C_Q^{oge}	0.0086	-0.0005	*-105.5%*	0.0004	*95.8%*
C_{My}^{oge}/C_Q^{oge}	-0.0033	-0.0007	*-78.6%*	-0.0014	*-58.9%*

Table 4
Rotor performance coefficients OGE $\mu = 0.05$.

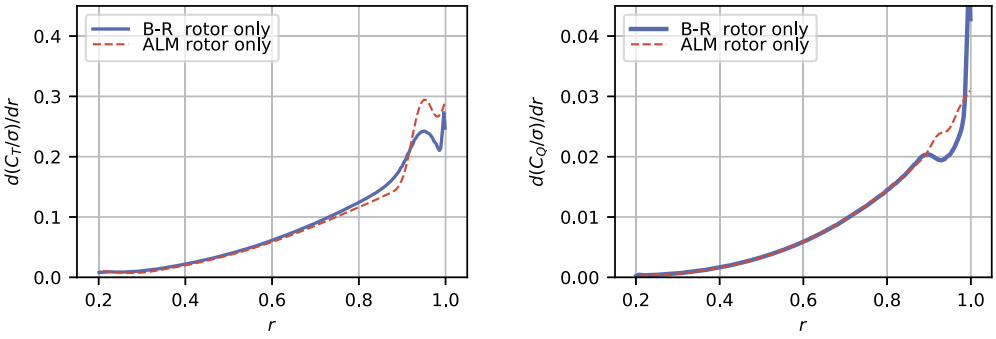
	Exp.	B-R	Δ	ALM	Δ
C_T^{oge}	7.550e-3	7.577e-3	0.4%	7.500e-3	-0.7%
C_Q^{oge}	7.626e-4	7.411e-4	-2.8%	7.596e-4	-0.4%
FoM^{oge}	0.608	0.629	3.5%	0.606	-0.4%
C_{Mx}^{oge}/C_Q^{oge}	-0.143	-0.625	*337.8%*	-0.668	*367.9%*
C_{My}^{oge}/C_Q^{oge}	1.212	0.996	*-17.9%*	0.745	*-38.5%*
C_{Mtot}^{oge}/C_Q^{oge}	1.221	1.176	-3.7%	1.001	-18.0%
$\Psi_{C_{Mtot}}^{oge}$	96.7^o	122.1^o	25.4^o	131.9^o	35.2^o

with an angle shift of 25° to 35° for the B-R and ALM respectively. This shift could be explained by the differences between the simplified numerical models and the more complex physical experimental setup. These values represent the reference conditions from which the remaining results in this work will be normalised.

4.1.1. Hover validation

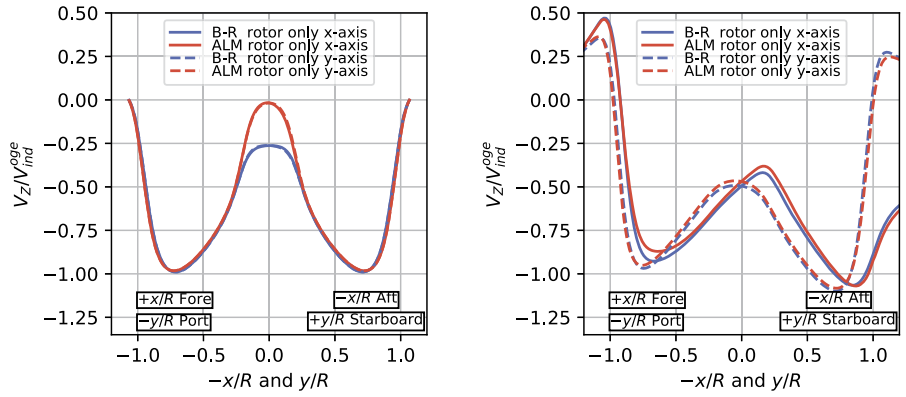
With the global performance coefficients of both numerical methods in good agreement with the experimental data, let us now consider a more detailed investigation of the two numerical methods compared to each other. This data is not present in the experimental database of Zagaglia et al. [53], but will serve as a more comprehensive validation of the actuator line method benchmarked against the fully blade resolved computations. The thrust and torque blade loading distributions are presented in Fig. 4 for the case in hover with no wind. The two methods are in excellent agreement in the first 80% to 90% of the blade span for both the thrust and torque distributions. Then, at 91.7% the thrust loading curves intersect, which coincides with the radial position of the tip vortex shed from the preceding blade, resulting in a stronger up and down-wash velocity for the ALM and therefore causing the discrepancy observed in terms of thrust loading. This behaviour has previously been observed in [41] due to the porous nature of the source terms used by the ALM. Despite this difference in loading near the tip, once integrated over the span, the difference in thrust and torque coefficients between the two methods corresponds to about 0.3% magnitude.

Next, the inflow velocity on a plane located 4 cm (5.33% D) above the rotor disk, normalised by their respective momentum theory hover-induced velocity V_{ind}^{oge}, is considered. The velocity is sampled on two lines corresponding to the x and y rotor axes and averaged over two full rotor revolutions. Positive x values correspond to the fore part of the helicopter whereas positive y values correspond to the starboard side. Results are presented in Fig. 5 for the two cases of hover and advancing flight out-of-ground-effect. In the wind-off case of Fig. 5a, the inflow distribution is symmetrical regardless of the axis or direction as expected from the almost nil pitching and rolling moments. From 30% blade span outwards, the two methods produce inflows that are almost identical to each other. In the centre of the rotor disk, the ALM produces a lower inflow than the B-R case due to a slightly weaker root vortex structure and the porous nature of the force. Moving to



(a) Thrust distribution (b) Torque distribution

Fig. 4. Blade loading distribution in hover.



(a) $\mu = 0.00$ (b) $\mu = 0.05$

Fig. 5. Inflow velocity along x and y axes 4 cm above rotor disk.

the wind-on case presented in Fig. 5b, the inflow distribution no longer is symmetrical. The addition of incoming wind modifies the flowfield around the rotor by reducing the induced velocity fore of the rotor centre (upwind) whereas it is increased in the aft direction (downwind). This longitudinal asymmetry is consistent with the experimental data presented in [65] and explains the strong positive (nose-up) pitching moment reported in Table 4 as a reduction in inflow velocity increases the effective angle of attack experienced by the blade in the fore position. Consequently, the thrust of this section will be larger and create the nose-up pitching moment. A similar inflow curve has also been experimentally measured by Taymourtash et al. [3] on the same helicopter model in similar operating conditions again with an associated strong nose-up pitching moment. Laterally, the counter-clockwise rotation of the rotor (viewed from the top) causes the advancing blade on the starboard side to see a greater freestream velocity, and therefore greater thrust, than on the port side. This lateral thrust imbalance causes the negative rolling moment towards the port side with the larger inflow magnitude on the starboard side being explained by the higher thrust on that part of the rotor.

Fig. 6 presents the tip vortex positions of the two methods averaged over their respective four rotor blades over 1.5 rotor revolutions in wake age. In addition, the empirical model of Kocurek and Tangler [66] is added for reference purposes. The curves at the top of the figure represent the radial contraction of the wake whereas the ones at the bottom are the axial advection of the tip vortices. Both methods are in excellent agreement with each other and the empirical model. The ALM tends to produce a wake that is ever so slightly more outboard than the B-R solution. Both methods experience some increased scattering in the positions of their tip vortices past 360° of wake age which is due to the degradation

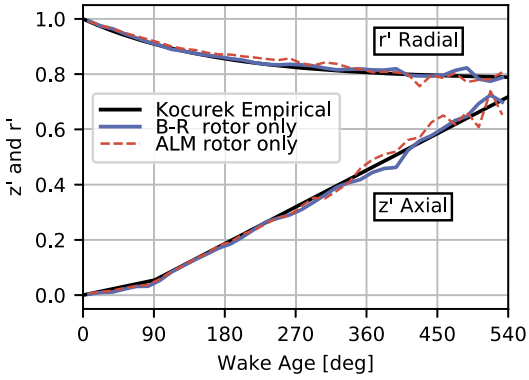


Fig. 6. Tip vortex trajectories in hover OGE.

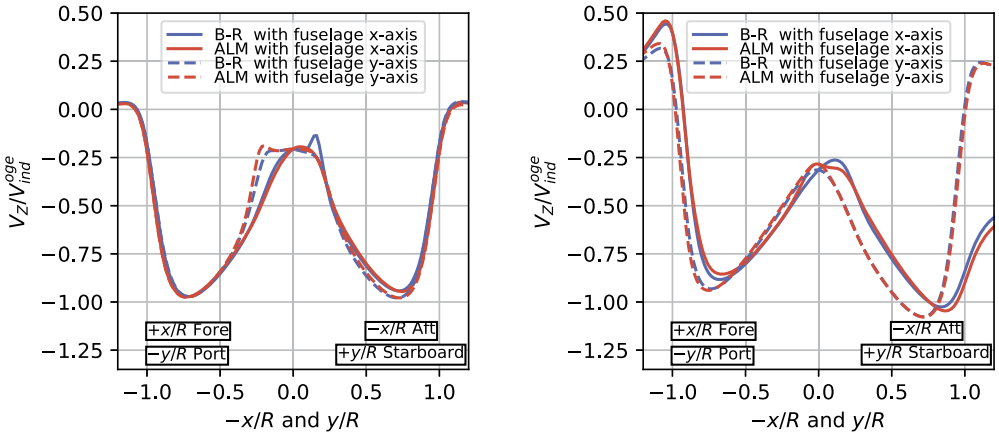
of the wake attributed to the dissipative nature of the employed mesh and numerical method. Note also that the turbulence model used did not use any rotational/curvature corrections, such as proposed by Shur et al. [67].

4.1.2. *Fuselage effects*

The experimental model featured a fuselage which, amongst other geometric features, has been neglected thus far in the simulations. The effects of the fuselage and the hub cover on the measured rotor loads are thus assessed in Table 5. In the wind-off case, the addition of the fuselage and hub cover slightly improve the thrust prediction of both methods when compared to the experimental data. However, the torque predictions for the B-R simulation is slightly degraded with an excess of torque which in turn

Table 5
Rotor performance coefficients with fuselage.

	$\mu = 0.00$				$\mu = 0.05$			
	B-R	Δ	**ALM**	Δ	**B-R**	Δ	**ALM**	Δ
C_T^{oge}	7.235E-3	-0.5%	7.268E-3	0.0%	7.679E-3	1.7%	7.533E-3	-0.2%
C_Q^{oge}	7.908E-4	1.3%	7.790E-4	-0.2%	7.662E-4	0.5%	7.613E-4	-0.2%
FoM^{oge}	0.550	-2.0%	0.562	0.2%	0.621	2.1%	0.608	0.0%
C_{Mx}^{oge}/C_Q^{oge}	0.0092	*7.1%*	0.0112	*30.5%*	-0.603	*322.5%*	-0.658	*360.7%*
C_{My}^{oge}/C_Q^{oge}	-0.0280	*747.9%*	-0.0173	*424.1%*	0.969	*-20.1%*	0.755	*-37.7%*
C_{Mtot}^{oge}/C_Q^{oge}	-	-	-	-	1.142	-6.5%	1.002	-18.0%
$\Psi_{C_{Mtot}}^{oge}$	-	-	-	-	121.9^o	25.2^o	131.1^o	34.3^o



(a) $\mu = 0.00$ (b) $\mu = 0.05$

Fig. 7. Inflow velocity along x and y axes 4 cm above rotor disk with fuselage.

causes a larger FoM error. The ALM on the other hand produces more consistent results with marginal errors. Both methods see their moment coefficients increase being on the order of 1% to 2% of the torque value which is more representative of the experimental roll moment, but too large for the pitching moment. Again, the relative errors are large due to the small value of the experimental data upon which they are normalised. For the wind-on case, a similar trend is present with an increase in both thrust and torque coefficient compared to the rotor only case. The moment coefficients do not show significant change for both numerical methods when considering the total moment coefficient and its relative angle. The ALM simulation once again shows less sensitivity to the inclusion of the fuselage and produces slightly better overall integrated results. Nevertheless, both methods are very close to the experimental data and comparable to the rotor only case.

The addition of the fuselage modifies the inflow distribution of both methods as shown in Fig. 7. In the wind off case, there is a clear asymmetry present, most noticeable near the rotor centre and the aft direction on the x-axis due to the tail boom geometry present in the area. The addition of the fuselage, and more specifically the rotor hub cover, negates the tendency of the ALM inflow near the centre to be underestimated compared to the B-R solution. With the presence of wind, the two methods see their distribution slightly modified compared with the rotor only case with most of the differences present near the centre for both axes and in the fore direction of the x-axis. The addition of the fuselage increases the agreement, in terms of overall inflow, between the two numerical methods along both axes in both wind conditions.

The average pressure coefficient, normalised by the hover induced velocity, on the fuselage top centre-line is presented in Fig. 8. For their respective wind condition, the two methods show

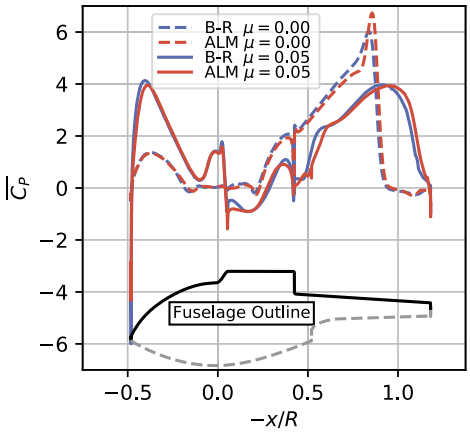
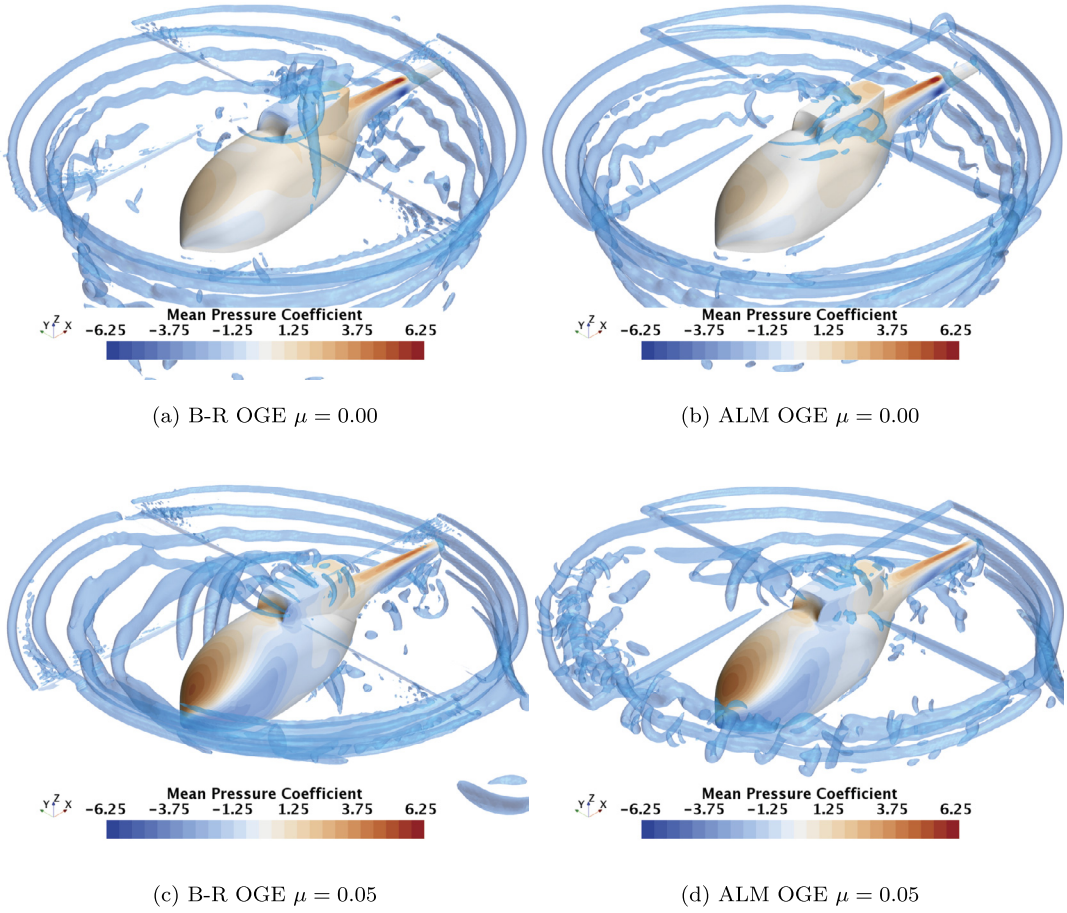


Fig. 8. Average pressure coefficient along fuselage top centre-line.

similar predictions with some portions of the pressure profiles being indistinguishable from each other. In the wind-off case, there is a marked influence of the rotor induced velocity on the tail boom culminating in a sharp increase in pressure coefficient at around 85% blade span which coincides with the position of the tip vortex. Adding wind skews the wake in the aft direction making both fore and aft tip vortices flow around the nose of the fuselage and the tip of the tail boom respectively. The increased pressure coefficient over the nose of the fuselage comes from the outboard section of the rotor induced velocity of the skewed wake. The tail boom receives an induced velocity more evenly distributed from the aft section of the rotor. Fig. 9 presents the surface pressure coefficient over the whole fuselage with a translucent Q-criterion



(a) B-R OGE $\mu = 0.00$ (b) ALM OGE $\mu = 0.00$

(c) B-R OGE $\mu = 0.05$ (d) ALM OGE $\mu = 0.05$

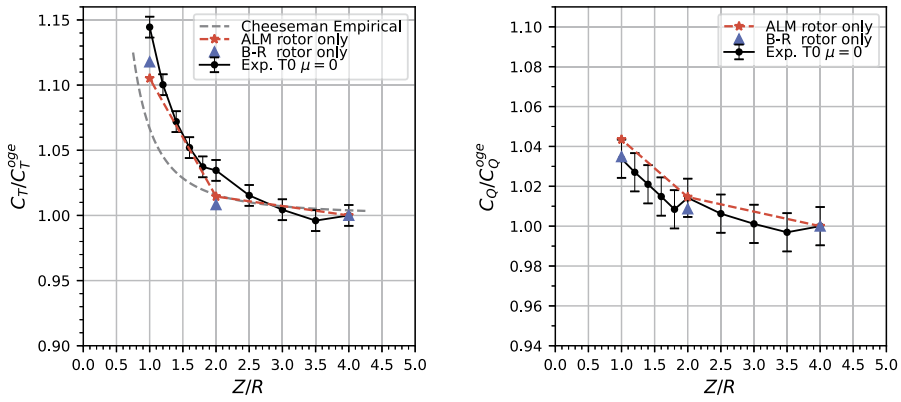
Fig. 9. Average surface pressure coefficient and Q-criterion contour at Q=75000 s^{-2}.

representation of the tip vortex wake. In addition to the aforementioned observations, we can observe a slight asymmetry between the starboard and port side of the fuselage due to the counter-clockwise rotation of the rotor. Most notably, the peak pressure coefficient near the nose is shifted slightly over the starboard side.

4.2. In-ground-effect - no obstacle

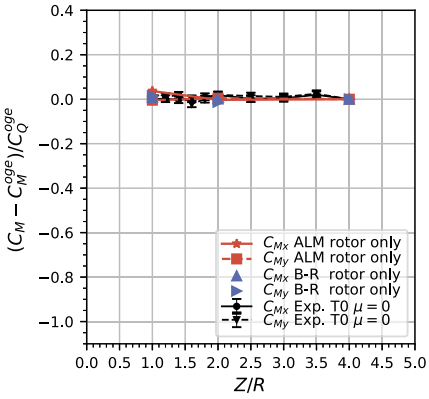
4.2.1. Hovering

In the experimental dataset [53], the ground effect in hover without the obstacle of the model scale helicopter is evaluated in Test 0 $\mu = 0.00$. Results are presented in Fig. 10 for the ALM, B-R and experimental data. Due to their high computational costs, the numerical methods are evaluated at only 3 heights corresponding to $Z/R = \{4.0, 2.0, 1.0\}$ with the highest corresponding to the reference OGE condition. The same plots as the original article are used with a preference to showcase the torque coefficient ratio instead of the figure of merit ratio. The FoM is dominated by the variation of the thrust and would yield an almost identical graph as the thrust ratio with a different scale. Therefore, both the thrust and torque coefficients of the helicopter rotor are normalised by their respective OGE values presented in Section 4.1 whereas the moment coefficient is presented as a difference to the OGE condition, normalised by the torque coefficient $(C_M - C_M^{oge})/C_Q^{oge}$. For this case, the thrust presented in Fig. 10a sees an increase related to the ground effect from $Z/R = 3.0$ for the experimental data whereas the computed points at $Z/R = 2.0$ also indicate a thrust increase for both methods that are more in line with the empirical model of Cheeseman and Bennett [68]. Decreasing the height to $Z/R = 1.0$ all data showcase a sharp rise in thrust as ground effect becomes more prevalent. The B-R and ALM solutions still slightly under-predict the thrust ratio by about 4% when compared to the experimental data whereas the Cheeseman model is lower by an additional 4%. It should be noted that the empirical model was derived from an assumption of constant power, which is not the present case. Overall, both numerical methods tend to underestimate the thrust coefficient rise when compared to the experimental data which has already been observed numerically by Gibertini et al. [69] in the case of an Actuator Disk model and Schmid [70] for an Unsteady Panel Method on the same geometry. The torque presented in Fig. 10b also presents an increase as the rotor is lowered with the two numerical methods also well capturing the trend and lying within the error bars of the experimental data. A similarly good agreement of the torque coefficient ratio was also found in Refs. [69,70] although their assessment of C_Q^{oge} was not as precise as the one in this work. The moment coefficients in both the rotor x and y-axis are presented in Fig. 10c where little variations are present for all cases. Moments corresponding to about 2-4% of C_Q^{oge} are present for the numerical methods at $Z/R = 1.0$ caused by a less stable wake topology of the root vortices present in-ground-effect causing asymmetry in the flowfield in particular near the rotor centre with a fountain-like effect causing unsteady upward velocities. Fig. 11 presents an open view of the tip-vortex wake topology and the instantaneous axial velocity in the X-Z plane. The two methods appear to be in excellent agreement both in terms of wake topology and axial velocity for a height of $Z/R = 4$. Close to the ground at $Z/R = 1$, the two methods produce a similar tip vortex wake topology in the outboard



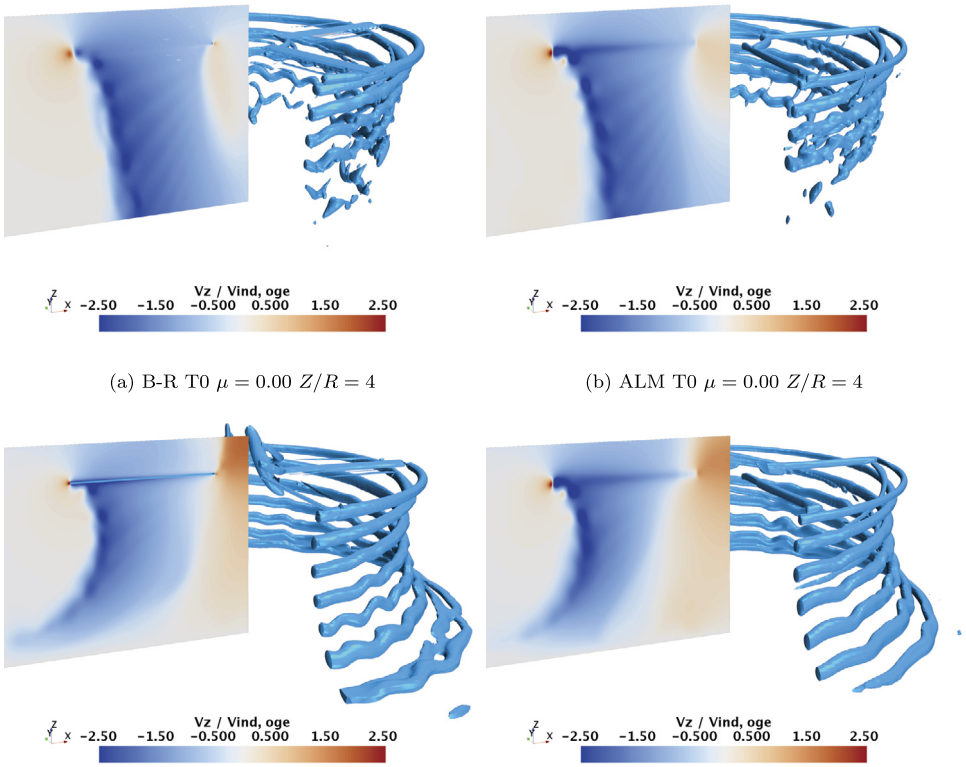
(a) Thrust coefficient ratio

(b) Torque coefficient ratio



(c) Moment coefficients variation

Fig. 10. Helicopter in IGE ($\mu = 0.00$) - performance coefficients comparison.



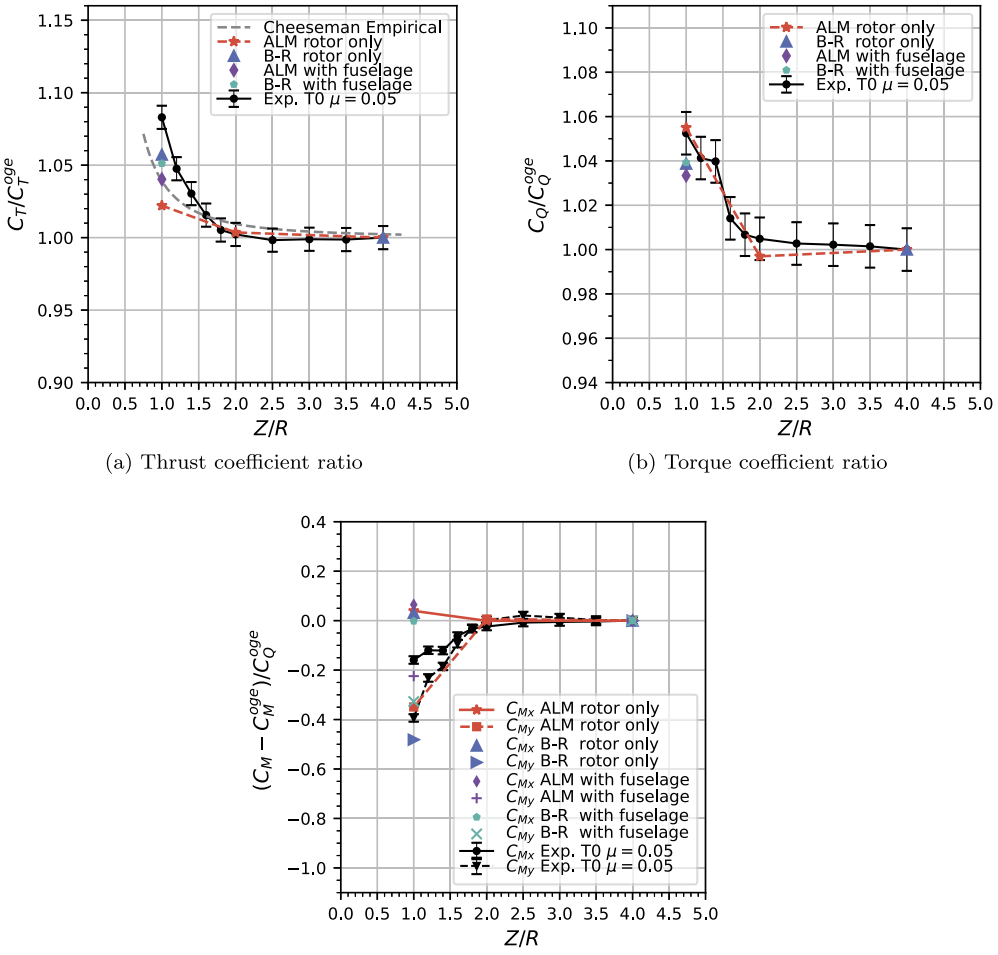
(a) B-R T0 $\mu = 0.00$ $Z/R = 4$

(b) ALM T0 $\mu = 0.00$ $Z/R = 4$

(c) B-R T0 $\mu = 0.00$ $Z/R = 1$

(d) ALM T0 $\mu = 0.00$ $Z/R = 1$

Fig. 11. Instantaneous axial velocity and Q-criterion contour at Q=25000 s^{-2}.



(a) Thrust coefficient ratio

(b) Torque coefficient ratio

(c) Moment coefficients variation

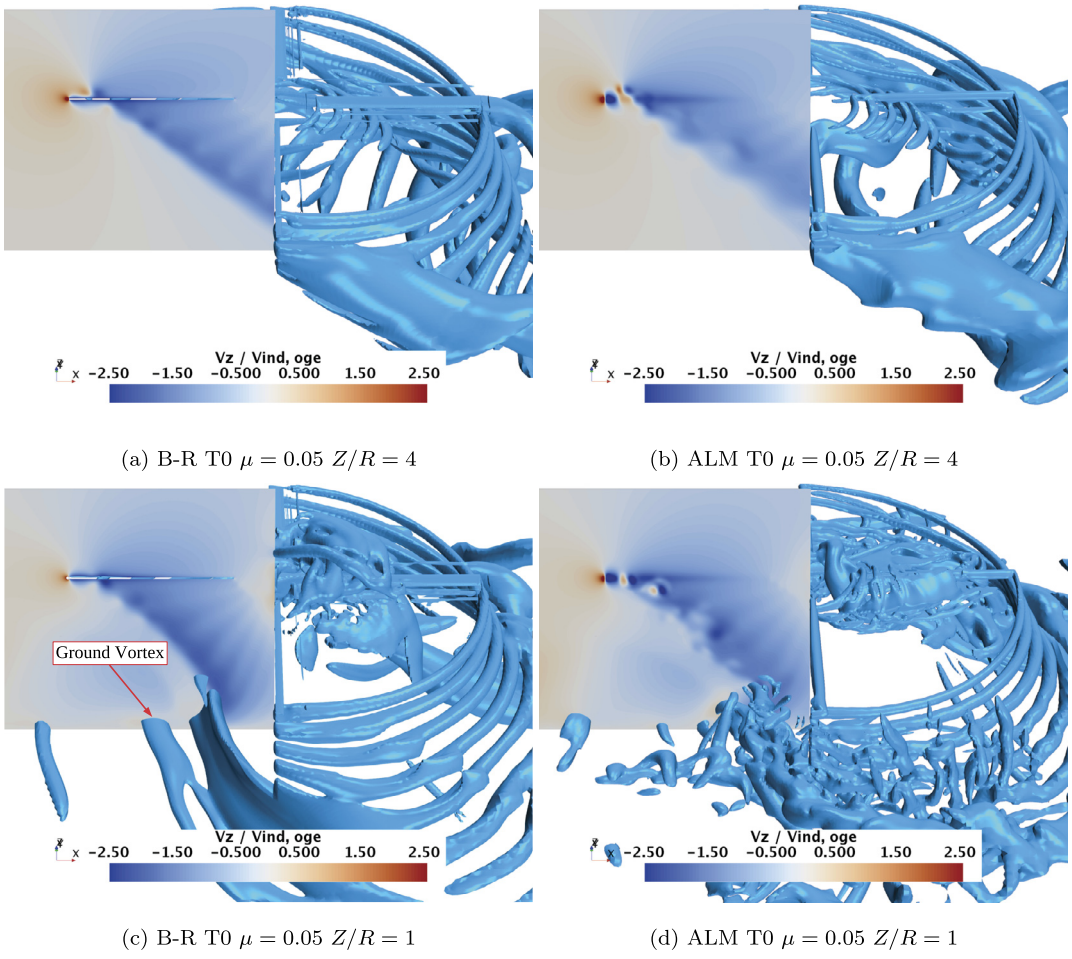
Fig. 12. Helicopter in IGE ($\mu = 0.05$) - performance coefficients comparison.

section of the rotor. However, the B-R produces a more constant downwash profile, especially near the rotor blade root. The resulting upward jet-like effect at the rotor centre is also stronger for the B-R case.

4.2.2. Advancing flight

As observed in Fig. 12, the presence of wind in-ground-effect delays the increase in thrust and torque coefficients observed in the previous section to rotor heights closer to the ground. The ALM captures well the slight thrust increase present at $Z/R = 2.0$, but underestimates the thrust value at $Z/R = 1.0$. An additional simulation with the fuselage geometry modelled is performed and improves the underestimated thrust to be more in-line with the Cheeseman and Bennett model as well as the B-R simulations (with and without the fuselage). The B-R solutions also underestimate the thrust ratio compared to the experimental data, but are in line with the empirical model and show less sensitivity to the addition of the fuselage in this case by reducing slightly the predicted thrust ratio. The torque coefficient ratio augmentation is again well captured with the rotor-only ALM despite the thrust coefficient ratio under-prediction. The other B-R and ALM simulations are also in fair agreement with values just below the experimental error bars. Looking at the moment coefficients in Fig. 12c, the negative difference in rolling moment compared to the OGE reference condition for the experimental data indicates an increase in the value as the reference is already negative. All the simulations predict a near-constant value with a small positive increase in the difference, thus a slight reduction compared to the OGE moment. The dominant pitching moment has a negative variation with its OGE value which translates to a reduction of its magnitude as the rotor is brought closer to the ground. The numerical methods in this case have their predictions forecasting the right trend, but experience more scatter. The addition of a fuselage has the effect of slightly diminishing the pitching moment reduction trend. Fig. 13 presents a qualitative assessment of the wakes for both methods. In OGE conditions depicted in Figs. 13a and 13b, the two methods produced near-identical wake structures with the typical flatten appearance and tip vortex roll-up into trailing supervortices. At $Z/R = 1$, we can observe the formation of a lingering ground vortex slightly upstream of the intersection of fore part of the skewed wake and the ground. This regime corresponds to the "ground vortex regime" identified by Brown and Whitehouse [27] at an identical rotor height above the ground and similar thrust normalised advance ratio. Their observations also suggest that this horseshoe-shaped ground vortex is less stable over the retreating blade region which is most noticeable for the ALM as illustrated in Fig. 13d.

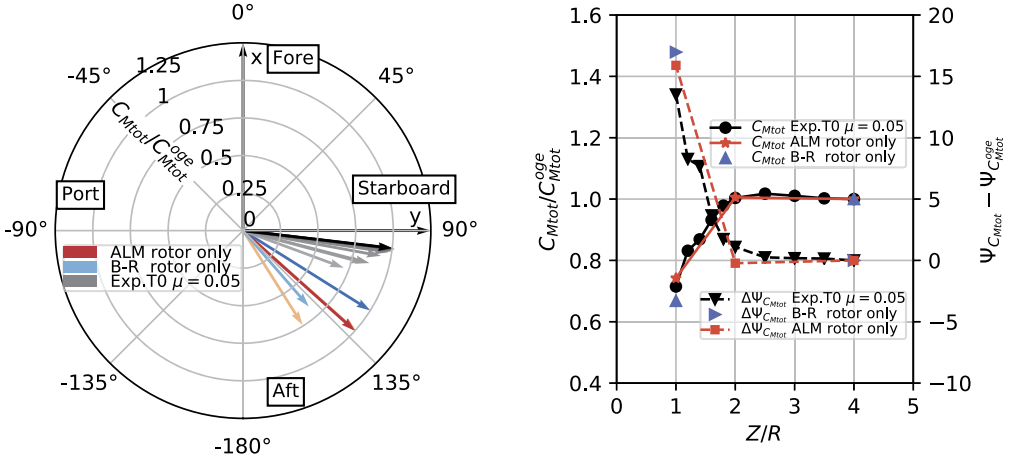
Although indicative of the performances of the rotor in terms of roll and pitch attitude, Fig. 12c does not correctly take into account the relative magnitude between the two moments and the offset between the experimental data and the simulations in OGE conditions reported in Section 4.1. A more insightful way of looking



(a) B-R T0 $\mu = 0.05$ $Z/R = 4$ (b) ALM T0 $\mu = 0.05$ $Z/R = 4$

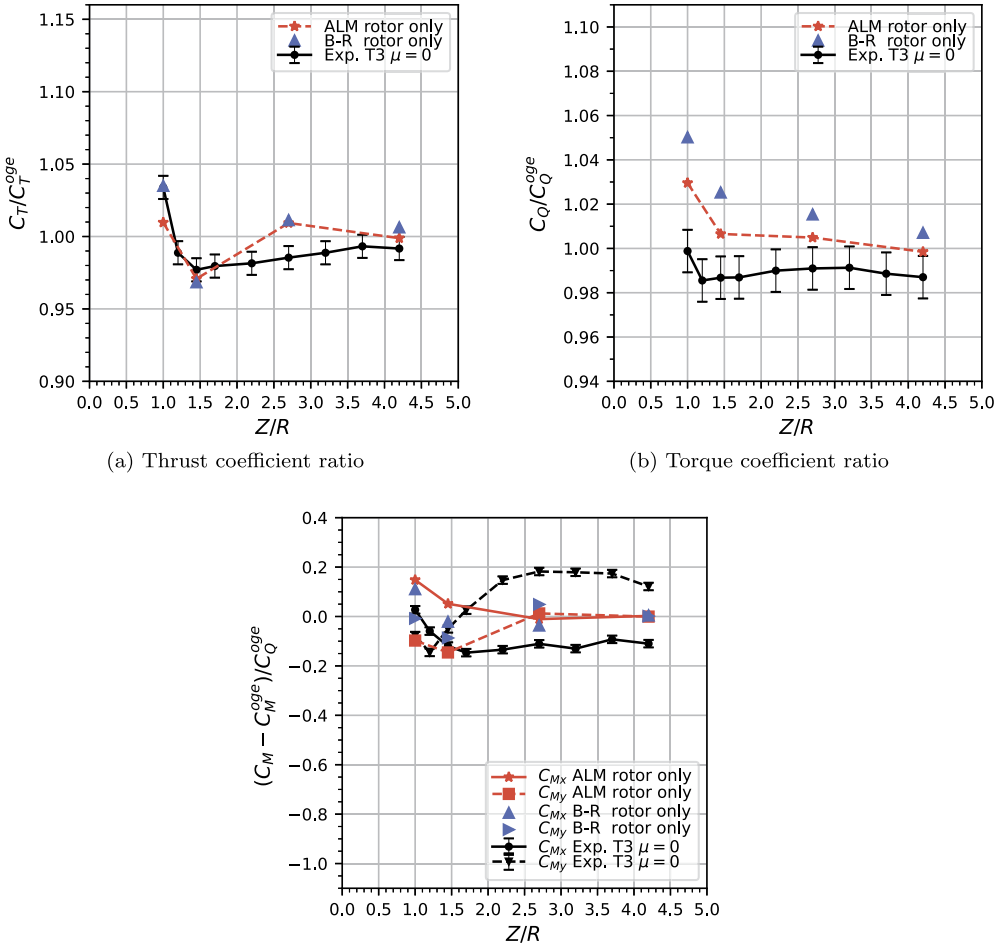
(c) B-R T0 $\mu = 0.05$ $Z/R = 1$ (d) ALM T0 $\mu = 0.05$ $Z/R = 1$

Fig. 13. Instantaneous axial velocity and Q-criterion contour at Q=7500 s^{-2}.



(a) Total moment coefficient ratio polar plot (b) Total moment coefficient ratio and orientation shift

Fig. 14. Helicopter in IGE $\mu = 0.05$ - total moment coefficient ratio and phase. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)



(a) Thrust coefficient ratio

(b) Torque coefficient ratio

(c) Moment coefficients variation

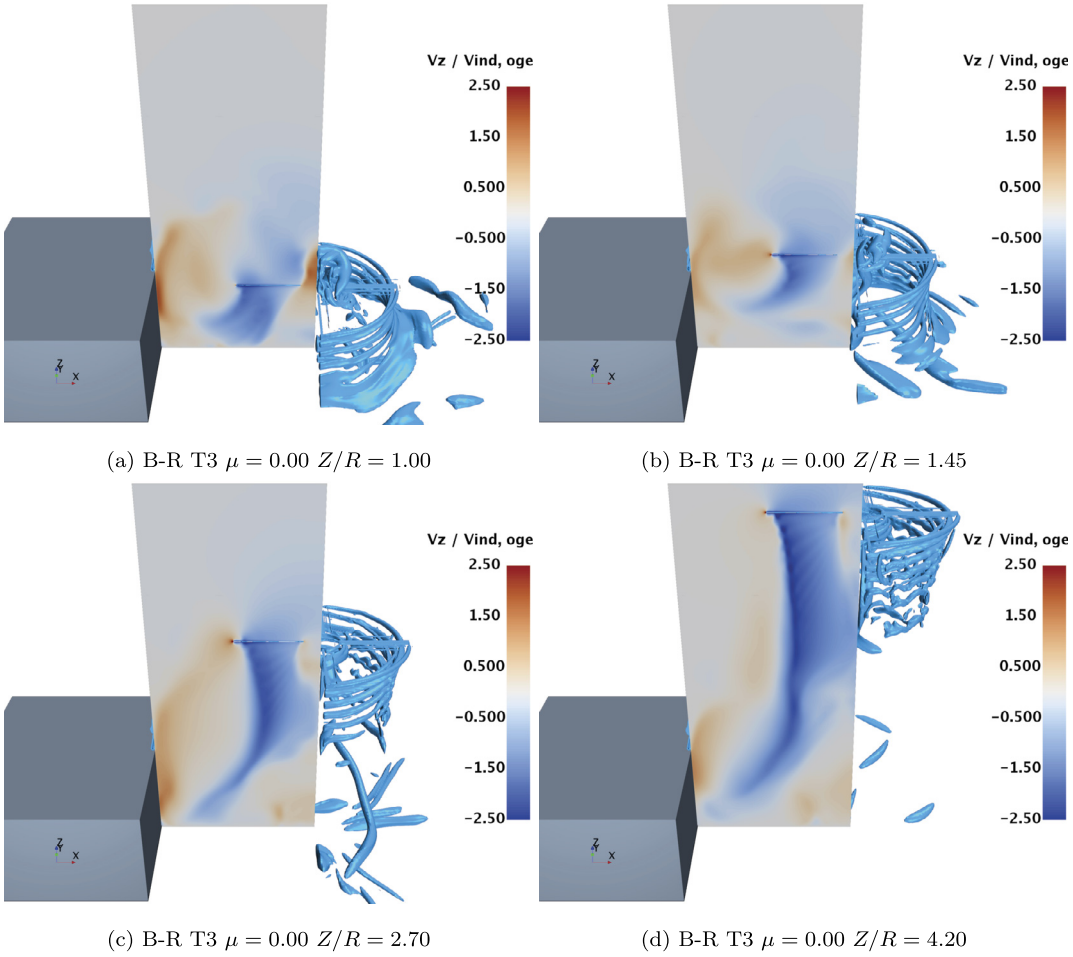
Fig. 15. Helicopter in IGE ($\mu = 0.00$) behind obstacle - performance coefficients comparison.

at the data is to compare the relative change of the total moment coefficient to its OGE reference value and the shift in its orientation. Fig. 14a presents a polar plot of the total moment relative magnitude and its orientation. Experimental data, B-R and the ALM results are presented in tones of grey, blue and orange respectively. The bolder/darker colours represent OGE conditions whereas the lighter tone gradient represents a rotor reducing its distance from the ground. To reduce clutter on the graphs, the rotor-only B-R and ALM simulations are presented. In all cases, a similar trend is observed for the reduction in total moment magnitude and shift in orientation across the simulated height range. Note that the polar plot moment at $Z/R = 2.0$ is indistinguishable from the OGE value as the ground effect does not yet play a major role at that height. Fig. 14b presents the numerical values of the moment ratio reduction and the orientation shift of the total moment with respect to the OGE condition. Both numerical methods correctly capture the overall trend accurately with a total moment reduction ratio of about 70% of the OGE value and orientation shift around 15°.

4.3. In-ground-effect - with obstacle

4.3.1. Hovering

The prismatic obstacle is added fore of the helicopter rotor, with the latter centred along the width of the obstacle, at a horizontal distance of $2R$ downstream. This configuration corresponds to Test 3 in the experimental database which features a sweep in the Z axis to evaluate the ground effect. Numerical results are computed at rotor heights of $Z/R = \{4.2, 2.7, 1.45, 1.0\}$ corresponding to experimental datapoints that showcased the largest variations. Both numerical methods consider the rotor-only configuration. Results are presented in Fig. 15. According to the experimental data, this configuration features a flow re-circulation region initiated by the ground and then the vertical wall of the obstacle streaming the rotor downwash upwards that is then re-ingested by the rotor as shown in Fig. 16. The flow re-circulation is most present and well-defined for rotor heights $Z/R \leq 1.5$ and causes a significant reduction in thrust in-ground-effect compared to the No-Obstacle case as observed in Fig. 15a. As the re-circulation is asymmetrical and is mostly present fore of the rotor, the increased inflow creates a local deficit in thrust which in turn becomes the observed nose-down pitching moment as explained in [53] and illustrated in Fig. 15c. These two values are well captured by the numerical methods for heights near the ground. In that region, the torque coefficient shows a consistent trend with the experimental data, but is slightly over-estimated, in particular for the blade resolved case. For heights $Z/R > 1.5$, the two numerical methods show very low variation when compared to the no obstacle case presented in Section 4.2.1 whereas the experimental data features a noted lower thrust and torque and the presence of both rolling and pitching moment on the order of 15% to 20% of C_Q^{oge} even when it is positioned at the maximum height of $Z/R = 4.2$. The presence of roll and pitch moments is probably due to a more complex re-circulation pattern that reaches and interacts with the rotor despite the significant height. As the experiment was run for a total



(a) B-R T3 $\mu = 0.00$ $Z/R = 1.00$ (b) B-R T3 $\mu = 0.00$ $Z/R = 1.45$

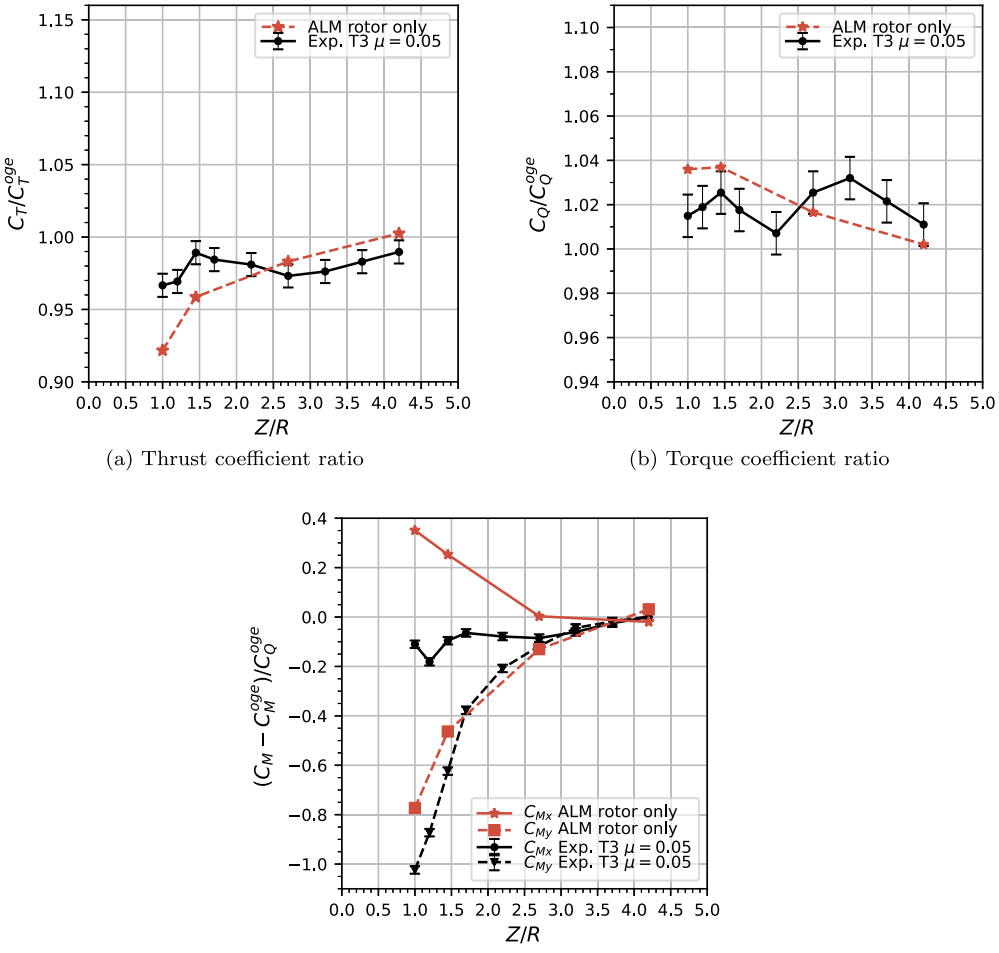
(c) B-R T3 $\mu = 0.00$ $Z/R = 2.70$ (d) B-R T3 $\mu = 0.00$ $Z/R = 4.20$

Fig. 16. Instantaneous axial velocity and Q-criterion contour at Q=7500 s^{-2}.

of 15 seconds (10 s flow stabilisation followed by 5 s data acquisition) which corresponds to a total of 645 rotor revolutions, we believe this flowfield is initiated over a much longer time period than the current simulation framework allows. The present simulations are run up to 75 to 100 revolutions until they are stopped due to excessive computational resources requirements of U-RANS based methods. The fully developed wakes of the simulations are represented in Fig. 16 where there is a clear re-circulation region that interacts with the rotor inflow for heights $Z/R \leq 1.5$ whereas for higher heights, the upward velocity region of the re-circulation is still located below the rotor disk and is not re-ingested by the rotor. The simulations at the higher heights are therefore stopped at about 75 revolutions as the rotor loading signals showed minimal or no variations with respect to OGE conditions. For reference, 75 revolutions corresponds to compute times of 9.27 days on 560 CPU cores for the B-R case whereas the ALM would require 4.6 days on the same amount of CPU cores. Reaching the full 675 revolutions as is done in the experiment would require a 9x increase in computational resources per data point which are already substantial.

4.3.2. In the presence of wind

Adding wind, the rotor is now placed in the wake of the obstacle. Due to the strong wake unsteadiness of the bluff body, an additional flow stabilisation period is simulated for 10 s at a larger time step of 0.01 s with only the obstacle modelled and the rotor removed from the simulation. The added simulation time allows a fully developed wake to form downstream of the obstacle where the rotor is then placed. For this configuration, only the ALM is considered due to the high computational resources required as the simulations are again run up to 100 revolutions. Performance coefficients are presented in Fig. 17. Contrary to the wind-off case presented in Section 4.3.1, the two points simulated above $Z/R = 2.0$ correctly represent the experimental data. In this case, there is less influence coming from the rotor-induced recirculating wake as the freestream and the obstacle wake dominate and transport the rotor wake downstream before it has a chance to re-circulate into the rotor. The flowfield is represented in Fig. 18, with the addition of line integral convolution, where the rotor at $Z/R = 2.70$ is positioned at the edge of the obstacle separated wake. The rotor at $Z/R = 4.20$ is positioned well above this region and is essentially subject to freestream conditions. Then, for heights $Z/R \leq 1.5$, the rotor is inside the detached wake of the obstacle which acts as a shield from the freestream for the rotor. As the rotor is brought closer to the ground, there is a decrease in thrust and a sharp drop in pitching moment that the ALM correctly captures when compared to the experimental data. This behaviour is explained by the formation of a rotor-induced re-circulation region similar to the wind-off case and illustrated in Figs. 18a and 18b. Although the variation of C_{Mx} of the experimental data and the ALM might seem to diverge in Fig. 17c, the two tend towards a similar value of $C_{Mx}/C_Q^{oge} = 0.25$ to 0.3 as the rotor height is lowered. The diverging trend is caused by the different OGE values reported previously. Finally, the moments are presented on a polar plot and in terms of ratio and orientation shift in Fig. 19. Again, the numerical method correctly captures



(a) Thrust coefficient ratio

(b) Torque coefficient ratio

(c) Moment coefficients variation

Fig. 17. Helicopter in IGE ($\mu = 0.05$) behind obstacle - performance coefficients comparison.

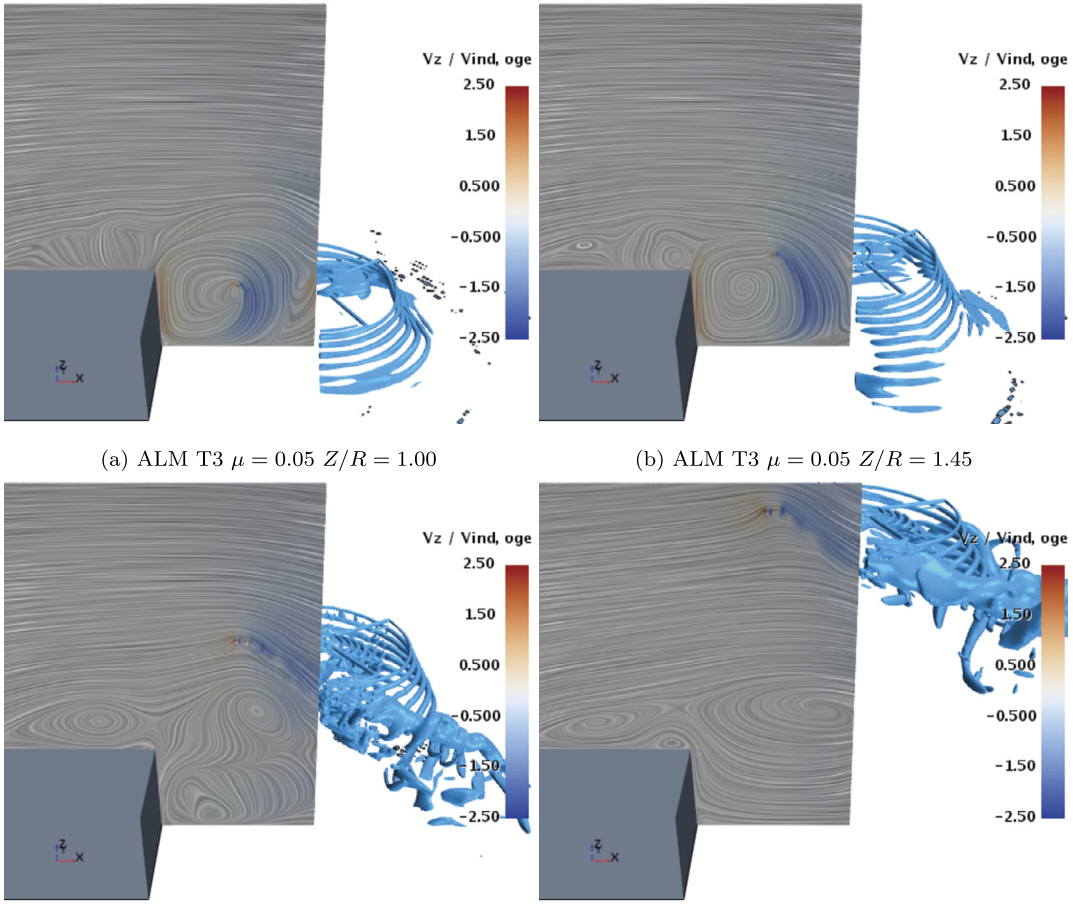
the trends and the magnitudes of the total moment ratio reduction and the orientation shift.

5. Conclusions

In this work, high fidelity CFD tools are used to assess their efficacy for the computation of a rotor in confined areas. Through a recently published experimental dataset, several key challenging helicopter flight conditions in ground effect including hovering, in forward flight, in the presence of a low-rise obstacle with and without the influence of wind are studied. Both fully Blade Resolved simulations and the Actuator Line Method are used as numerical methods and are benchmarked against each other. The principal observations are:

- Outside of ground effect, the two numerical methods accurately predict the main rotor coefficients with the exception of the moments, most notably the rolling moment, when compared to experimental data. Between them, the agreement of the ALM and B-R solutions over performance coefficients, blade loading, wake geometry, inflow and surface pressure coefficients are excellent.
- Through the reduction in mesh size from the removal of the rotor geometry, the ALM requires less computational resources than the B-R. For the present rotor geometry and test cases, the computational cost of an ALM simulation is about half of a B-R case and is directly proportional to the total mesh cell count.
- Hovering in-ground-effect, the two methods correctly capture the trends of the experimental data with a slight under-prediction of thrust whereas the torque and both moments are accurately captured.
- For the advancing flight case in-ground-effect, a similar trend is present as the thrust in-ground-effect is under-predicted and the torque increase is well captured. Accounting for the mismatch in rolling and pitching moments in OGE conditions, the variation in total moment magnitude ratio and orientation shift is well captured.
- The addition of an obstacle for a hovering rotor causes flow recirculation that is captured for rotors placed near the ground. Simulations with rotors placed higher could not achieve a sufficiently developed flowfield for the given computational resources and therefore did not accurately capture a proper behaviour in terms of thrust and moments.
- The presence of wind with the ground obstacle is well captured overall, however, thrust is under-predicted for rotors placed near the ground.

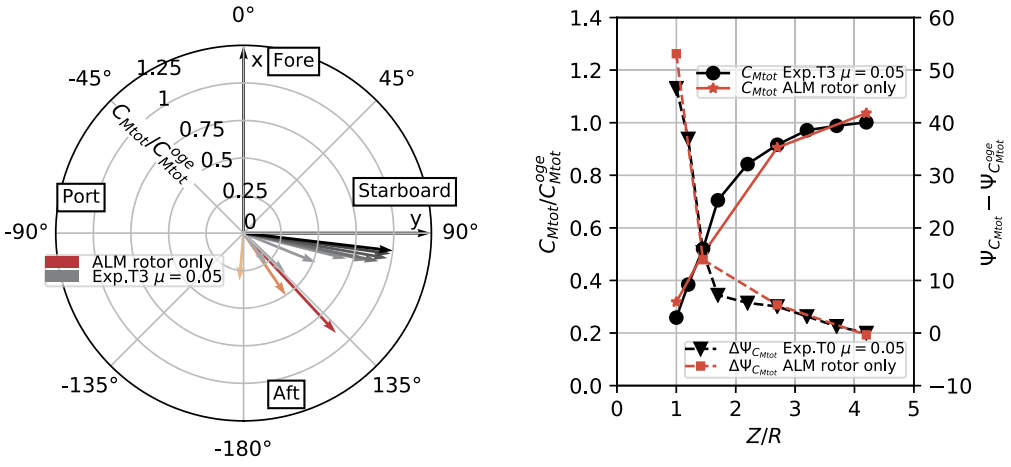
In conclusion, the ALM appears to be a suitable alternative to fully resolved CFD simulations at a fraction of the computational cost making the source term based method attractive for studying complex aerodynamic configurations and cases with a higher number of evaluated data-points. The ALM correctly predicts the overall trends and response magnitudes in challenging flight conditions when compared to blade resolved simulations. Further numerical studies with the ALM should be aimed at incorporating more



(a) ALM T3 $\mu = 0.05$ $Z/R = 1.00$ (b) ALM T3 $\mu = 0.05$ $Z/R = 1.45$

(c) ALM T3 $\mu = 0.05$ $Z/R = 2.70$ (d) ALM T3 $\mu = 0.05$ $Z/R = 4.20$

Fig. 18. Instantaneous axial velocity with total velocity vector superimposed through line integral convolution and Q-criterion contour at Q=7500 s^{-2}.



(a) Total moment coefficient ratio polar plot (b) Total moment coefficient ratio and orientation shift

Fig. 19. Helicopter in IGE ($\mu = 0.05$) behind obstacle - total moment coefficient ratio and phase.

complex aero-mechanical movements, trim procedures and blade structural dynamics solvers for the evaluation of more realistic helicopter rotors.

CRediT authorship contribution statement

Reda Merabet: Methodology, Software, Visualisation, Simulation, Writing.

Eric Laurendeau: Supervision, Writing Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] A. Hoencamp, T. van Holten, J.V.R. Prasad, Relevant Aspects of Helicopter-Ship Operations, in: Proceedings of the 34th European Rotorcraft Forum ERF34, Vol. 1, Royal Aeronautical Society, Liverpool, United Kingdom, 2008, pp. 578–588.

[2] R.G. Lee, S.J. Zan, Wind tunnel testing of a helicopter fuselage and rotor in a ship airwake, J. Am. Helicopter Soc. 50 (4) (2005) 326–337, https://doi.org/10.4050/1.3092869.

[3] N. Taymourtash, D. Zagaglia, A. Zanotti, G. Gibertini, G. Quaranta, Wind Tunnel Investigation of a Helicopter Model in Shipboard Operations, Warsaw, Poland, 2019, pp. 1–13.

[4] N. Taymourtash, V. Muscarello, G. Quaranta, Gust and Inflow Model Identification from Wind Tunnel Tests for Rotorcraft-Obstacle Interaction, Rome, Italy, 2019, pp. 1981–1989.

[5] N. Taymourtash, D. Zagaglia, A. Zanotti, V. Muscarello, G. Gibertini, G. Quaranta, Experimental study of a helicopter model in shipboard operations, Aerosp. Sci. Technol. (2021) 106774, https://doi.org/10.1016/j.ast.2021.106774.

[6] C. Crozon, R. Steijl, G.N. Barakos, Numerical study of helicopter rotors in a ship airwake, J. Aircr. 51 (6) (2014) 1813–1832, https://doi.org/10.2514/1.C032535.

[7] I. Oruc, J.F. Horn, S. Polsky, J. Shipman, J. Erwin, Coupled flight dynamics and CFD simulations of the helicopter / ship dynamic interface, in: American Helicopter Society 71st Annual Forum, AHS International, Virginia Beach, Virginia, 2015.

[8] Y. Shi, G. Li, D. Su, G. Xu, Numerical investigation on the ship/multi-helicopter dynamic interface, Aerosp. Sci. Technol. 106 (2020) 106175, https://doi.org/10.1016/j.ast.2020.106175.

[9] Y. Shi, D. Su, G. Xu, Numerical investigation of the influence of passive/active flow control on ship/helicopter dynamic interface, Aerosp. Sci. Technol. 106 (2020) 106205, https://doi.org/10.1016/j.ast.2020.106205.

[10] J.F. Tan, T.Y. Zhou, Y.M. Sun, G.N. Barakos, Numerical investigation of the aerodynamic interaction between a tiltrotor and a tandem rotor during shipboard operations, Aerosp. Sci. Technol. 87 (2019) 62–72, https://doi.org/10.1016/j.ast.2019.02.005.

[11] A. Visingardi, F.D. Gregorio, T. Schwarz, M. Schmid, R. Bakker, S. Voutsinas, Q. Gallas, R. Boisard, G. Gibertini, D. Zagaglia, G. Barakos, R. Green, G. Chirico, M. Giuni, Forces on Obstacles in Rotor Wake – A GARTEUR Action Group, Milan, Italy, 2017, pp. 1–14.

[12] Q. Gallas, R. Boisard, J.-C. Monnier, J. Pruvost, A. Gilliot, Experimental and Numerical Investigation of the Aerodynamic Interactions Between a Hovering Helicopter and Surrounding Obstacles, Milan, Italy, 2017, pp. 1–6.

[13] D. Zagaglia, M. Giuni, R.B. Green, Rotor-obstacle aerodynamic interaction in hovering flight: an experimental survey, in: American Helicopter Society 72nd Annual Forum, AHS International, West Palm Beach, Florida, 2016.

[14] D. Zagaglia, G. Gibertini, M. Giuni, R.B. Green, Experiments on the Helicopter-Obstacle Aerodynamic Interference in Absence of External Wind, Lille France, 2016.

[15] D. Zagaglia, M. Giuni, R.B. Green, Investigation of the rotor–obstacle aerodynamic interaction in hovering flight, J. Am. Helicopter Soc. 63 (3) (2018) 1–12, https://doi.org/10.4050/JAHS.63.032007.

[16] T. Jardin, S. Prothin, C.G. Magaña, Aerodynamic performance of a hovering microrotor in confined environment, J. Am. Helicopter Soc. 62 (2) (2017) 1–7, https://doi.org/10.4050/JAHS.62.022008.

[17] S. Prothin, C. Fernandez Escudero, N. Doué, T. Jardin, Aerodynamics of MAV rotors in ground and corner effect, Int. J. Micro Air Veh. 11 (2019) 1–13, https://doi.org/10.1177/1756829319861596.

[18] W. Johnson, Helicopter Theory, Dover Publications, New York, 1994.

[19] R. Quackenbush, K. Chua, D. A. Wachspress, H. Boschitsch, Computation of Rotor Aerodynamic Loads in Forward Flight Using a Full-Span Free Wake Analysis, NASA CR 177611, NASA, Oct. 1990.

[20] C. Ferlisi, Rotor Wake Modelling Using the Vortex-Lattice Method, Master's thesis, Ecole Polytechnique Montreal, Montreal, Canada, Apr. 2018.

[21] J. Katz, A. Plotkin, Low-Speed Aerodynamics, 2nd edition, Cambridge University Press, New York, 2001.

[22] D.A. Griffiths, S. Ananthan, J.G. Leishman, Predictions of rotor performance in ground effect using a free-vortex wake model, J. Am. Helicopter Soc. 50 (4) (2005) 302–314, https://doi.org/10.4050/1.3092867.

[23] J.F. Tan, J.G. Cai, G.N. Barakos, C. Wang, M.Q. Huang, Computational study on the aerodynamic interference between tandem rotors and nearby obstacles, J. Aircr. 57 (3) (2020) 456–468, https://doi.org/10.2514/1.C035629.

[24] D.-G. Caprace, P. Chatelain, G. Winckelmans, Lifting line with various mollifications: theory and application to an elliptical wing, AIAA J. 57 (1) (2019) 17–28, https://doi.org/10.2514/1.J057487.

[25] D.-G. Caprace, P. Chatelain, G. Winckelmans, Wakes of rotorcraft in advancing flight: a large-eddy simulation study, Phys. Fluids 32 (8) (2020) 087107, https://doi.org/10.1063/5.0015162.

[26] T.S. Kalra, V.K. Lakshminarayan, J.D. Baeder, CFD Validation of Micro Hovering Rotor in Ground Effect, AHS International, Phoenix, Arizona, 2010, pp. 1–22.

[27] R.E. Brown, G.R. Whitehouse, Modelling rotor wakes in ground effect, J. Am. Helicopter Soc. 49 (3) (2004) 238–249, https://doi.org/10.4050/JAHS.49.238.

[28] R.G. Rajagopalan, S.R. Mathur, Three dimensional analysis of a rotor in forward flight, J. Am. Helicopter Soc. 38 (3) (1993) 14–25, https://doi.org/10.4050/JAHS.38.14.

[29] F. Le Chuiton, Actuator disc modelling for helicopter rotors, Aerosp. Sci. Technol. 8 (4) (2004) 285–297, https://doi.org/10.1016/j.ast.2003.10.004.

[30] W. Khier, Time-accurate versus actuator disk simulations of complete helicopters, in: W.E. Nagel, M. Resch, W. Jäger (Eds.), High Performance Computing in Science and Engineering' 05, Springer-Verlag, Berlin/Heidelberg, 2006, pp. 209–220.

[31] D.M. O'Brien Jr, M.J. Smith, Analysis of rotor-fuselage interactions using various rotor models, in: 43rd AIAA Aerosp. Sci. Meet., vol. 468, 2005.

[32] Y. Shi, Y. Xu, K. Zong, G. Xu, An investigation of coupling ship/rotor flowfield using steady and unsteady rotor methods, Eng. Appl. Comput. Fluid Mech. 11 (1) (2017) 417–434, https://doi.org/10.1080/19942060.2017.1308272.

[33] D.D. Boyd Jr, Rotor/fuselage unsteady interactional aerodynamics: a new computational model, PhD Thesis, Virginia Tech, 1999.

[34] G. Barakos, T. Fitzgibbon, A. Kusyumov, S. Kusyumov, S. Mikhailov, CFD simulation of helicopter rotor flow based on unsteady actuator disk model, Chin. J. Aeronaut. 33 (9) (2020) 2313–2328, https://doi.org/10.1016/j.cja.2020.03.021.

[35] I. Oruc, J.F. Horn, J. Shipman, Coupled flight dynamics and computational fluid dynamics simulations of rotorcraft/terrain interactions, J. Aircr. 54 (6) (2017) 2228–2241, https://doi.org/10.2514/1.C034101.

[36] G. Chirico, D. Szubert, L. Vigevano, G.N. Barakos, Numerical modelling of the aerodynamic interference between helicopter and ground obstacles, CEAS Aeronaut. J. 8 (4) (2017) 589–611, https://doi.org/10.1007/s13272-017-0259-y.

[37] J.N. Sørensen, W.Z. Shen, Numerical modeling of wind turbine wakes, J. Fluids Eng. 124 (2) (2002) 393–399, https://doi.org/10.1115/1.1471361.

[38] E. Alpman, L.N. Long, D.O. Bridges, J.F. Horn, Fully-coupled simulations of the rotorcraft / ship dynamic interface, in: American Helicopter Society International 63rd Annual Forum, AHS International, Virginia Beach, Virginia, 2007, pp. 1367–1382.

[39] J.R. Forsythe, E. Lynch, S. Polsky, P. Spalart, Coupled flight simulator and CFD calculations of ship airwake using kestrel, in: 53rd AIAA Aerospace Sciences Meeting, American Institute of Aeronautics and Astronautics, 2015.

[40] Y. Delorme, R. Stanly, S.H. Frankel, D. Greenblatt, Application of actuator line model for large Eddy simulation of rotor noise control, Aerosp. Sci. Technol. 108 (2021) 106405, https://doi.org/10.1016/j.ast.2020.106405.

[41] R. Merabet, E. Laurendeau, Hovering helicopter rotors modeling using the actuator line method, J. Aircr. 59 (3) (2022) 774–787, https://doi.org/10.2514/1.C036314.

[42] W.Z. Shen, J.H. Zhang, J.N. Sørensen, The actuator surface model: a new Navier–Stokes based model for rotor computations, J. Sol. Energy Eng. 131 (1) (2009) 011002, https://doi.org/10.1115/1.3027502.

[43] C. Sibuet Watters, C. Masson, Modeling of lifting-device aerodynamics using the actuator surface concept, Int. J. Numer. Methods Fluids 62 (2009) 1264–1298, https://doi.org/10.1002/fld.2064.

[44] C. Sibuet Watters, S. Breton, C. Masson, Application of the actuator surface concept to wind turbine rotor aerodynamics, Wind Energy 13 (5) (2009) 433–447, https://doi.org/10.1002/we.365.

[45] C. Masson, C.S. Watters, Moving actuator surfaces: a new concept for wind turbine aerodynamic analysis, in: Proceedings of the International Conference on Renewable Energies and Power Quality (ICREPQ08), Santander, Spain, 2008.
[46] D. Linton, R. Widjaja, B. Thornber, Validation of an Actuator Surface Model with CFD-Convected Wake Model for Hover and Forward Flight, Jeju Island, Korea, 2018, pp. 1–8.
[47] D. Linton, R. Widjaja, B. Thornber, Simulations of Tandem and Coaxial Rotors Using a CFD-Coupled Rotor Model, Adelaide, Australia, 2018, pp. 1–4.
[48] D. Linton, B. Thornber, An actuator surface method for ship-helicopter dynamic interface simulations, in: AIAA Scitech 2021 Forum, American Institute of Aeronautics and Astronautics, Virtual Event, 2021.
[49] S.J. Zan, Experimental determination of rotor thrust in a ship airwake, J. Am. Helicopter Soc. 47 (2) (2002) 100–108, https://doi.org/10.4050/JAHS.47.100.
[50] R.G. Lee, S.J. Zan, Unsteady aerodynamic loading on a helicopter fuselage in a ship airwake, J. Am. Helicopter Soc. 49 (2) (2004) 149–159, https://doi.org/10.4050/JAHS.49.149.
[51] C.H. Kääriä, Y. Wang, G.D. Padfield, J.S. Forrest, I. Owen, Aerodynamic loading characteristics of a model-scale helicopter in a ship's airwake, J. Aircr. 49 (5) (2012) 1271–1278, https://doi.org/10.2514/1.C031535.
[52] C.H. Kääriä, Y. Wang, M.D. White, I. Owen, An experimental technique for evaluating the aerodynamic impact of ship superstructures on helicopter operations, Ocean Eng. 61 (2013) 97–108, https://doi.org/10.1016/j.oceaneng.2012.12.052.
[53] D. Zagaglia, A. Zanotti, G. Gibertini, Analysis of the loads acting on the rotor of a helicopter model close to an obstacle in moderate windy conditions, Aerosp. Sci. Technol. 78 (2018) 580–592, https://doi.org/10.1016/j.ast.2018.05.019.
[54] STAR-CCM+ Documentation, User Manual Version 12.06, Siemens PLM Software, 2016.
[55] J.H. Ferziger, M. Perić, Computational Methods for Fluid Dynamics, 3rd edition, Springer, Berlin; New York, 2002.
[56] F.R. Menter, Two-equation eddy-viscosity turbulence models for engineering applications, AIAA J. 32 (8) (1994) 269–289, https://doi.org/10.2514/3.12149.
[57] M.J. Churchfield, S.J. Schreck, L.A. Martinez, C. Meneveau, P.R. Spalart, An advanced actuator line method for wind energy applications and beyond, in: 35th Wind Energy Symposium, American Institute of Aeronautics and Astronautics, 2017.
[58] L.A. Martinez-Tossas, M.J. Churchfield, C. Meneveau, Optimal smoothing length scale for actuator line models of wind turbine blades based on Gaussian body force distribution, Wind Energy 20 (6) (2017) 1083–1096, https://doi.org/10.1002/we.2081.
[59] M. Shives, C. Crawford, Mesh and load distribution requirements for actuator line CFD simulations, Wind Energy 16 (8) (2012) 1183–1196, https://doi.org/10.1002/we.1546.
[60] L.A. Martinez-Tossas, C. Meneveau, Filtered lifting line theory and application to the actuator line model, J. Fluid Mech. 863 (2019) 269–292, https://doi.org/10.1017/jfm.2018.994.
[61] R. Merabet, E. Laurendeau, Parametric study on the velocity sampling techniques for the actuator line method in 2D, in: AIAA SciTech 2019 Forum, American Institute of Aeronautics and Astronautics, San Diego, California, 2019.
[62] R. Martinuzzi, C. Tropea, The flow around surface-mounted, prismatic obstacles placed in a fully developed channel flow (data bank contribution), J. Fluids Eng. 115 (1) (1993) 85–92, https://doi.org/10.1115/1.2910118.
[63] J.-Y. Hwang, K.-S. Yang, Numerical study of vortical structures around a wall-mounted cubic obstacle in channel flow, Phys. Fluids 16 (7) (2004) 2382–2394, https://doi.org/10.1063/1.1736675.
[64] H.C. Lim, T. Thomas, I.P. Castro, Flow around a cube in a turbulent boundary layer: LES and experiment, J. Wind Eng. Ind. Aerodyn. 97 (2) (2009) 96–109, https://doi.org/10.1016/j.jweia.2009.01.001.
[65] J. W. Elliott, S. L. Althoff, Inflow measurement made with a laser velocimeter on a helicopter model in forward flight. Volume I: Rectangular Planform blades at an advance ratio of 0.15, NASA TM 100541, National Aeronautics and Space Administration, Apr. 1988.
[66] J.D. Kocurek, J.L. Tangler, A prescribed wake lifting surface hover performance analysis, J. Am. Helicopter Soc. 22 (1) (1977) 24–35, https://doi.org/10.4050/JAHS.22.24.
[67] M.L. Shur, M.K. Strelets, A.K. Travin, P.R. Spalart, Turbulence modeling in rotating and curved channels: assessing the spalart-shur correction, AIAA J. 38 (5) (2000) 784–792, https://doi.org/10.2514/2.1058.
[68] I.C. Cheeseman, W.E. Bennett, The Effect of the Ground on a Helicopter Rotor in Forward Flight, A.R.C. Technical Report 3021, Aeronautical Research Council, London UK, Sep. 1955.
[69] G. Gibertini, G. Droandi, D. Zagaglia, P. Antoniazza, A.O. Catelan, CFD Assessment of the Helicopter and Ground Obstacles Aerodynamic Interference, Lille France, 2016, pp. 1–13.
[70] M. Schmid, Simulation of Helicopter Aerodynamics in the Vicinity of an Obstacle Using a Free Wake Panel Method, Milan, Italy, 2017, pp. 1–14.